An Evaluative Study on IoT Ecosystem for Smart Predictive Maintenance (IoT-SPM) in Manufacturing: Multiview Requirements and Data Quality

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***Abstract*—With the recent advances of the Internet of Things (IoT), innovative techniques, and concepts have emerged, such as digital twins and industrial 4.0. As one of the essential parts of a digital twin, IoT-based smart predictive maintenance (IoT-SPM) is a key enabling technology for smart manufacturing. This article introduces digital twins and their relationship to IoT-SPM and proposes a reference IoT-SPM, aiming to provide a comprehensive and systematic outlook for the IoT-SPM field. Thus, it can be used as a guide map for interested readers. To give a complete picture of the IoT-SPM ecosystem in industrial 4.0 systems, this article conducts an analysis from multiview perspectives, starting with the architecture, followed by platforms and component. The key components or requirements of an IoT-SPM ecosystem are identified and outlined, including the IoT and cyber–physical system (CPS) as the cornerstone technologies, IoT monitoring data as the base, big data platforms as the backbone, an upgraded computing paradigm as the catalyst, and machine learning-based data analysis as the main processor. This article also focuses on the issues surrounding IoT data when applying analytic models to a real-world industrial IoT system. Then, the current progresses relating IoT and IoT-SPM are depicted, and a research gap on IoT data quality is identified. In particular, regarding the identified IoT data quality problems, this article qualitatively evaluates and discusses the existing solutions. These discussions lead to several open research issues and future directions.**

***Index Terms*—Data quality, ecosystem requirements, Internet of Things (IoT), reference IoT ecosystem, smart manufacturing.**

I. Introduction

DIGITAL twins, which are precise, virtual copies of machines or systems, are revolutionizing industry [1]. Although the concept was introduced in 2002 by Michael Grieves, it was not well discussed or developed until recent years since information was limited and mostly manually collected back then. With the advent of the Internet of Things (IoT) and big data (BD), more digital twins have become a reality. Information and communication technology (ICT) are enabling keys for realizing the transformation, making digital twins efficient, and responsive. Driven by the BD collected from IoT sensors in real time, digital twins mirror almost every facet of a product, process or service and can be used to control its real-world counterpart. Digital twins are entering mainstream use and many companies and governments already use them. For example, NASA uses digital twins to monitor spacecraft [2]. New South Wales is creating a digital twin of the state to facilitate better planning, design, and modeling for future needs [3]. Singapore developed a digital twin for the entire city [4]. One analyst predicts half of all corporations might be using digital twins by 2021 [5].

Digital twins are being deployed in various fields [6], [7], [8] and offer opportunities for smart cities [9], smart manufacturing, smart healthcare, smart traffic, smart farming [10], and so on. For example, in comparison to traditional cities, smart cities are more efficient and sustainable with the use of information, digital, and telecommunication technologies to provide a better quality of life for its citizens and visitors, particularly with the unprecedented challenges of continuous urbanization and increasing population density. Smart manufacturing aims to convert data acquired across the product lifecycle into manufacturing intelligence to positively impact all aspects of manufacturing [11]. Particularly in industry, digital twins have been also implemented in different sectors and applications [12], [13], [14], [15], [16], such as design, production, manufacturing, and maintenance. IoT-based smart predictive maintenance (IoT-SPM) is a key part of smart manufacturing in Industry 4.0 systems and one of the most researched applications in digital twins field [17], [18], [19], [20]. Machines are fitted with sensors, actuators, and Wi-Fi capabilities to track vibrations, temperature, speed, power, and so on through an integrated network. The collected real-time data can be used for various smart factory applications, including digital twins and predictive maintenance, enabling the manufacturer to make informed decisions in a timely manner.

IoT-SPM can assist the manufacturer in numerous ways as follows.

1) *Reduce Maintenance Costs:* The primary purpose of an IoT-SPM solution is to predict equipment failure for

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predictive maintenance. The often used "fail and fix" and the periodic check approach to ensure machine reliability is not an effective way to measure the health of new and complex digitized factory equipment. IoT-SPM takes monitored sensor data into account, analyzing not only the component's present status but also predicts future problems and when they are likely to occur. IoT-SPM enables engineers to carry out maintenance only when a fault is detected early, thus it can greatly reduce maintenance costs.

2) *Extend the Lifetime of Machines:* IoT-SPM allows manufacturers to monitor the health state condition of equipment in a real-time manner. The early detection and maintenance of possible faults that might occur in a piece of equipment enables collateral failures to be prevented in other equipment that is connected to it. Issues, such as wear-and-tear, overheating, and other threats to the health of equipment can be resolved in a timely manner, thereby lengthening the lifespan of a machine. Sometimes collateral failures can be catastrophic but scheduled maintenance can potentially prevent catastrophic failures.
3) *Maximise Production Output:* To enable predictive maintenance, IoT-SPM can minimize unplanned downtime and provide engineers with a substantial time window to schedule an inspection. This, in turn, reduces any associated disruption of the whole production chain with significant consequential cost savings.
4) *Optimize Inventory Management:* If we can detect a fault and identify which component has caused the fault in advance based on IoT-SPM, the results can be used to optimize the spare part inventory to avoid running out of spares or overstocking on expensive spare parts and avoid logistical delays. Spare parts are often designed and made through special processes, and sometimes they might need to be imported from overseas.

To create a digital twin in smart manufacturing for the purpose of the IoT-SPM ecosystem, BD is generated by multisource and densely installed IoT sensors. In addition, as an IoT data-driven ecosystem, data quality directly influences the monitoring performance. Considering these issues, IoT architecture-assisted solutions are used, where BD platforms, cloud and fog computing, and machine learning tools, can be strategically leveraged and integrated in support of intelligent IoT-SPM ecosystems.

Although several studies on IoT have been conducted on various aspects, including IoT architecture [21], [22], IoT data collection [23], IoT integration [24], IoT data storage [25], [26], IoT security [27], [28], IoT data analysis [29], and IoT data in the cloud [30], [31], none gives a complete picture of the IoT-SPM ecosystem. Motivated by this phenomenon, this article specifically focuses on exploring every component throughout the IoT-SPM ecosystem. With the aim of providing a more detailed reference architecture along with implementation guidelines, this article reports a more comprehensive discussion in terms of ecosystem multiview requirements, current progress, and identified gaps in data quality. From the perspective of architecture, the ecosystem generally requires three layers, namely, device layer, network layer, and application layer. From the perspective of components, IoT and cyber–physical system (CPS) technologies, IoT monitoring data, BD platforms, computing resource distribution, and machine learning-driven data mining techniques are the most essential parts. Fig. 1 presents the structure outline of this article. Specifically, the contributions of this article are listed as follows.

1) The key requirements for an IoT-SPM ecosystem are identified and outlined from the perspectives of architecture and components.
2) We investigate the development history of IoT and the characteristics and challenges during the process from collecting, transforming to analyzing IoT monitoring data.
3) As the backbone to support the whole ecosystem, application programming interface-oriented BD platforms are discussed and categorized.
4) To boost the processing capabilities, we present the need for the upgrades in terms of computing architecture to accommodate stricter timeliness guarantees, such as fog, edge, and cloud computing.
5) The most essential part of the ecosystem is IoT-related monitoring data. Most IoT architecture-related papers are reported from academic perspectives, which lack the involvement of industry experts. This article identifies the difficulties and data quality issues that might be encountered in the data application layer especially for an IoT-SPM ecosystem.
6) This article provides an outlook for the current progress of IoT and existing IoT-SPM ecosystems and a comprehensive qualitative evaluation on the solutions on data quality issues especially for IoT applications.

II. IoT-SPM Ecosystem: Multiview Requirements

The basic idea of the IoT is the pervasive presence around us of a variety of things or objects, such as radio-frequency identification tags (RFIDs), sensors, actuators, mobile phones, and so on [32]. By enabling easy access and interaction with a wide variety of such devices, the IoT will foster the innovative development of a large number of applications in many different domains, including smart transportation, smart home, smart health, smart farms, smart manufacturing industries, etc. The enormous amount and variety of data generated from the IoT enable smart applications to provide new services to citizens, companies, and public administrations. Various researchers have described the IoT in a multitude of forms.

1) The term IoT was initially proposed to refer to uniquely identifiable interoperable connected objects with the RFID technology [33].
2) Later on, researchers related the IoT with more objects, such as RFID tags, sensors, actuators, GPS devices, mobile devices, and other devices which have the ability to sense and transmit data from the physical world to a virtual system.
3) Today, the definition of the IoT is accepted in an even boarder sense. It refers to a dynamic global network

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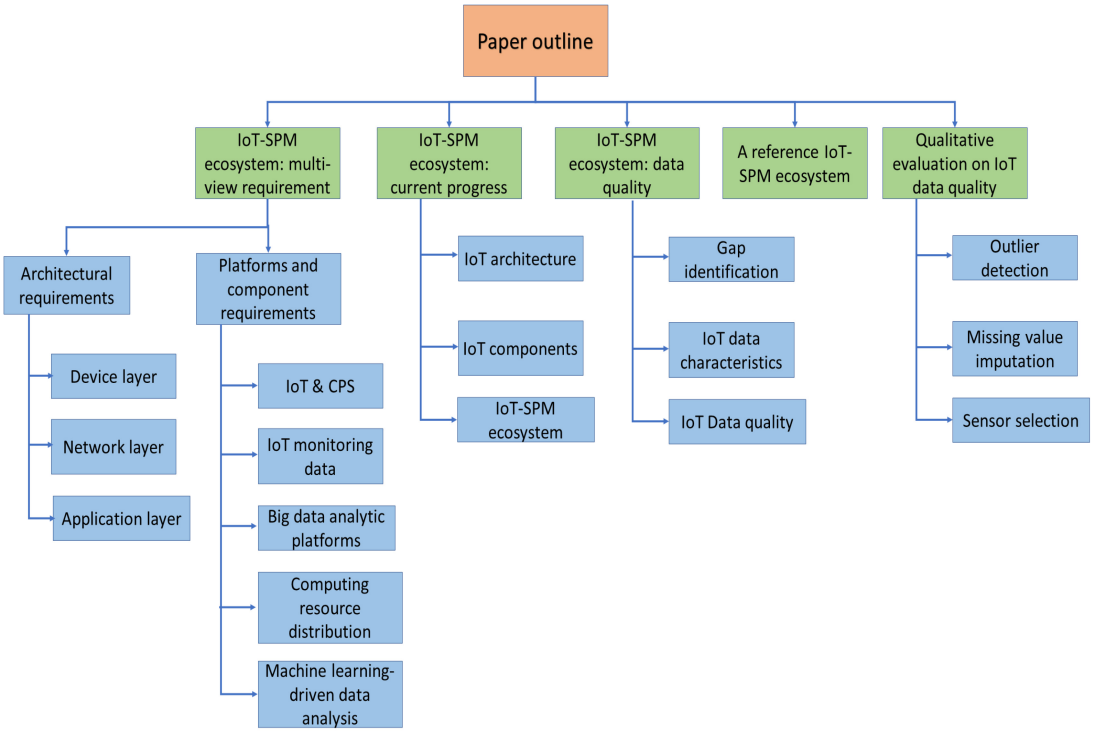


Fig. 1. Structure outline of this article.

infrastructure with self-configuring capabilities based on standard and interpretable communication protocols where physical and virtual "things" have identities, physical attributes, and virtual personalities and use intelligent interfaces, and are seamlessly integrated into the information network [34].

4) Alternatively, paper [35] defines the IoT as a recent communication paradigm that envisions a near future, in which the objects of everyday life will be equipped with microcontrollers, transceivers for digital communication, and suitable protocol stacks that will enable them to communicate with one another and with the users, becoming an integral part of the Internet.
5) Another description is given in [36], where the IoT is the network of physical objects, devices, vehicles, buildings, and other items which are embedded with electronics, software, sensors, and network connectivity, enabling these objects to gather and exchange data.

The IoT is rapidly gaining ground as a novel paradigm in the scenario of modern wireless telecommunications. A commonly accepted definition of the IoT is a whole global infrastructure enabling advanced services by interconnecting physical and virtual things using interoperable ICTs. Specifically, the IoT comprises three parts: 1) the things (objects); 2) the communication networks that connect them; and 3) the computer systems using data streaming from and to objects [36]. The evolution of IoT is illustrated in Fig. 2.

The actualization of IoT concepts in the real world is possible through several enabling technologies. It is realized by building a complete ecosystem that makes use of these technologies to convert the operation data generated by IoT into informative analysis results. In particular, the exploding

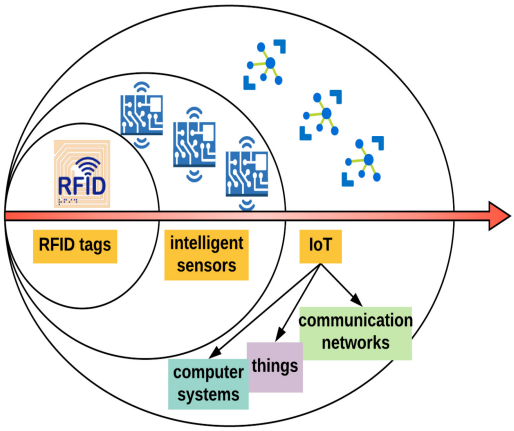


Fig. 2. Evolution of IoT.

growth of connected devices from heterogeneous sources and the diversity of network technologies have yielded concerns regarding scalability and sustainability. A promising solution lies in the integrated ecosystem which provides a foundation for connected devices to collect, manage, and process the IoT sensor data by the embedded scalable tools and algorithms. In an IoT ecosystem, it is requisite to manage BD properly from heterogeneous devices so as to conduct data analytics to get the desired output. In this regard, developers can focus on the added value of a particular application. The main enabling factors of such an ecosystem are the integration of enabling technologies and communication solutions. As a paradigm that is envisioned to involve all the smart things to be connected on the Internet, the IoT has extensible and compatible features and is able to integrate diverse interdisciplinary technologies and absorb their advantages from every aspect of the information

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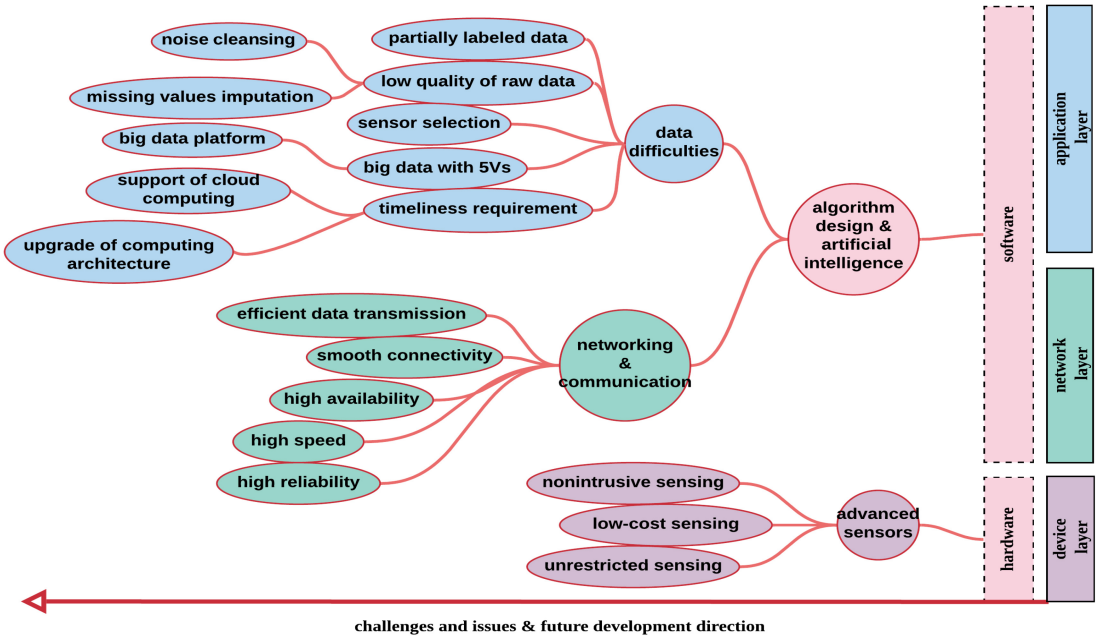


Fig. 3. Potential issues and future directions.

domain. In this section, we introduce several IoT-related background concepts, and provide a comprehensive investigation of each of the enabling factors from the perspectives of architecture and components. On the one hand, five components are required in support of intelligent IoT-SPM ecosystems, including IoT monitoring data as the source of information and knowledge, IoT and CPS technologies as the cornerstone to enable the implementation, BD platforms as the backbone to support the IoT BD analysis, new computing paradigms as the catalyst to speed up the ecosystem process, and machine learning tools as the main processor to dig out the high value hidden in the IoT source data. On the other hand, for an IoT-SPM, an IoT architecture is a key enabler that strategically integrates and leverages the aforementioned five components to enhance IoT-SPM effectiveness.

A. Architecture Requirements

IoT is identified as a disruptive technology to realize smart manufacturing with the use of IoT architecture-based digital twins. IoT architecture generally consists of three tiers: 1) sensory devices; 2) network for communications; and 3) data management and analysis, and is divided into three layers: 1) the perception (or device) layer; 2) the network layer; and 3) the application layer [37]. The work in [21] comprehensively investigates state-of-the-art research on IoT architecture, and outlines the key requirements for future works to accommodate the anticipated trillions of devices which are expected to be connected to the Internet in the foreseeable future.

Low-level sensory devices in the perception layer are responsible for sensing and gathering data from physical entities, including the surrounding environmental conditions and monitoring areas of interest. A continuum of devices is connected to the Internet through heterogeneous access network technologies, such as RFID and wireless sensor networks (WSNs). The network layer receives the information sent by the perception layer and transmits it to the application layer via connected networks. In this layer, the data are transferred through interfaces or gateways among the heterogeneous networks and use a variety of communication technologies and protocols, such as Bluetooth, WiFi, ZigBee, and so on. Based on the collected data, the application layer is where engineers or data scientists extract useful information and hidden values from the IoT sensor data.

The aforementioned three-layer IoT architecture is a complete process from sensing objects in a physical space, to analyzing the information in cyber space. The physical hardware entities in the device layer are closely correlated with the networking setup in the network layer and the virtual optimized software algorithms in the application layer under a unified architecture (UA). To develop a more powerful IoT architecture, major changes to both the hardware and software level are required. Fig. 3 summarises the potential challenges and issues that need to be considered in the three layers of an IoT architecture from the aspects of hardware and software. For researchers of interest, the work in [38] and [21] provides more detailed and in-depth investigations and discussions and could be good references.

1) *Device Layer:* Although advances in modern sensor technologies enable the sensibility of more metrics, such as temperature and humidity, vibration and noise, rotation speed, liquid leakage, etc., their sensing capabilities still need to be extended and improved. For example, intrusive sensors might require hardware updates when adding dedicated sensors into the existing equipment. In addition, to ensure seamless monitoring at all possible points, low-cost sensing is desirable to make it affordable to deploy high-density networked sensors.
2) *Network Layer:* To meet the demand resulting from an increasing number of sensor devices, high speed, reliability, and availability in terms of network connection

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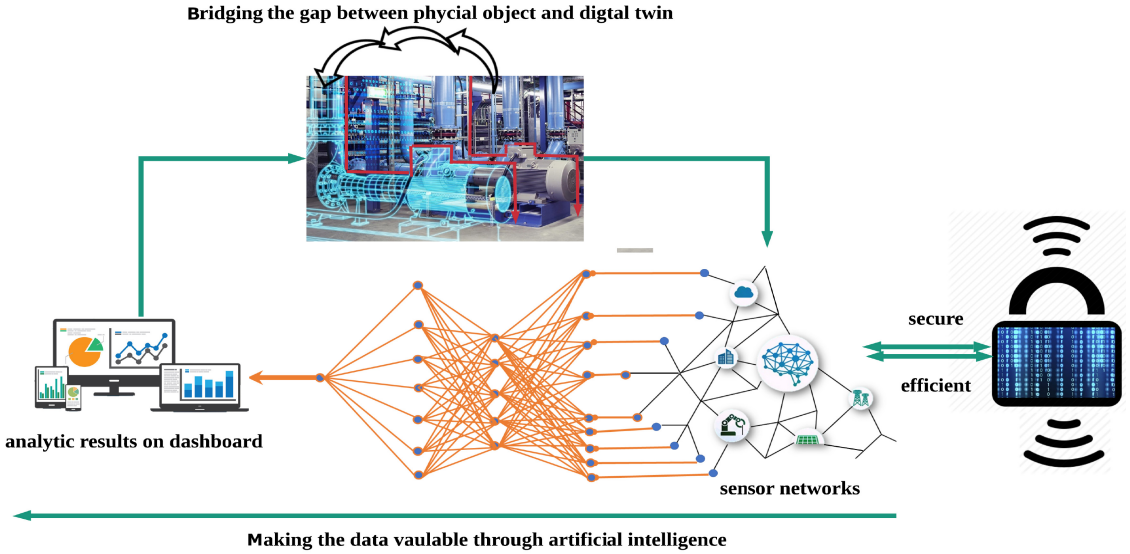


Fig. 4. Physical object and its digital twin in smart manufacturing.

are prerequisites for smooth connectivity. The diversity in IoT devices with different specifications and standards requires the interoperability between them. Thus, new forms of networking and communication are anticipated for efficient and secure data transmission.

3) *Application Layer:* The application layer typically performs machine learning-driven data analysis to offer appropriate suggestions to decision makers. However, traditional machine learning methods cannot be simply copied into the IoT application layer. They have to be modified to be capable of real-time processing and an immediate response and they also must be compatible with cloud-assisted IoT architecture.

B. Platforms and Component Requirements

To implement IoT concepts in industry practice, experts from both industry and academic domains are focused on developing enabling technologies and have proposed the concept of Industry 4.0. Over the last few years, the smart manufacturing industry has undergone an ever-enriching process. In addition to IoT, many other modern technologies have been implemented, including CPS, BD platforms, cloud computing, and artificial intelligence. Although these concepts have been in existence in the context of ICT for several years, they have recently been applied to industrial automation, aiming to change the way smart manufacturing works in modern industry.

1) IoT and CPS: CPS manage interconnected systems between physical assets and computing systems. Through CPS, physical objects, and computing nodes are able to connect and cooperate with each other via various means of communications. Briefly, CPS is the intersection of physical and cyber systems. The work in [39] emphasises that the smart interconnection between the cyber and physical space is the foundation of CPS. With the greater availability and affordability of smart sensors, sensing technologies, data management systems, and computer networks, CPS is widely recognized as enabling smart manufacturing [39]. IoT technologies provide opportunities to digitalize physical machines with unique sensing capabilities, and CPS enables the efficient interaction between the physical world and its digital counterpart. Evolving from IoT and CPS, respectively, Industrial IoT (IIoT) and ICPS are defined specifically for industrial applications.

Integrating IIoT and ICPS led to the revolution of automation technology in the industrial 4.0 era [40], [41]. Initially proposed in Germany, the Industry 4.0 concept has gained visibility and attention on a global scale. IIoT allows the digital descriptions of physical machines, while ICPS interconnects them in the physical space and computing capabilities in the cyber space by smart interaction. By enabling cooperation and communication, digital information can be transferred, stored, and updated in a real-time manner in Industry 4.0 systems, and represents a second identity of the machine itself which is a sort of digital twin. As illustrated in Fig. 4, physical objects can be digitalized using the real-time data collected from sensor networks, then a digital counterpart can be built based on these data. Once a digital twin is developed, the inside data is analyzed using artificial intelligence technologies, such as deep learning and the analytic results can be delivered to engineers in a streaming manner. In this way, decision makers are able to remotely monitor the physical object based on its digital twin in a centralized office. It is noted that with the rapid development and adaptation of IoT technologies, the concept of IoT has been extended. The massive adoption of Internet protocol (IP)-enabled sensory devices in CPS and the increasingly powerful wireless connectivity has blurred the boundary between CPS and the IoT [42]. Further discussions can also be found in [42].

2) IoT Monitoring Data: Most manufacturing processes can be described either by precise mechanism models based on expert knowledge or by BD-driven models. In traditional industry practice, knowledge-based approaches provide most solutions to real-time monitoring [43] which is the main factor of smart manufacturing. Faced with the complexity induced

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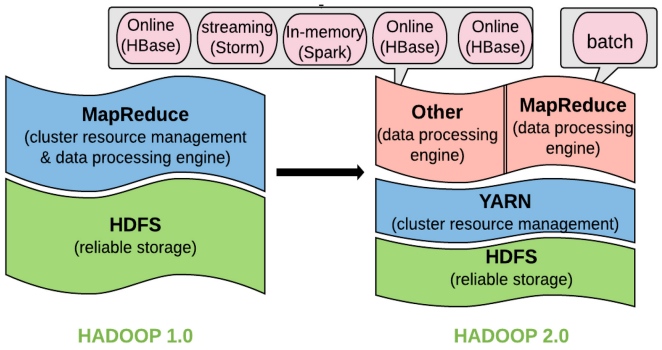


Fig. 5. Components of a Hadoop stack and its development history.

by recent large-scale hierarchical and distributed systems, process knowledge is not always available and the estimation of mathematical models becomes increasingly unpractical. With the rapid adaptation of smart sensor technologies, IoT-enabled solutions have recently gained much attention in industry and manufacturing. Intensive networked sensors provide seamless monitoring for the manufacturing process, and their sensing capabilities allow the remote checking of the process condition through the collected monitoring data. In such a case, data-driven approaches that are based on the monitoring historical data and online measured data are of particular interest.

Manufacturing data is the foundation of conducting data-driven approaches. One BD solution cannot fit all manufacturing circumstances since different applications are associated with different data types, formats, and characteristics, thus specific tools and algorithms are required to solve these issues. Therefore, systematically analyzing the raw data as the first step helps us understand what is involved in each task and provides useful guidelines on choosing the appropriate BD-enabling technologies to develop an optimized solution for this specific task.

3) Big Data Analytic Platforms: Although IoT BD brings deep insights into smart manufacturing, developing analytic platforms encounters nontrivial challenges due to the complex nature of BD. To deal with this, researchers explore possible solutions and there are two common approaches, which are called vertical and horizontal scaling in the work in [44] and are also known as scale up and scale out, respectively, in the work in [45]. Vertical scaling methods boost the power of platform with the use of a new larger computer system (in terms of memory and speed) to replace the old one. Horizontal scaling distributes the overall workload into multiple independent computing machines, in which case tasks can be processed in a parallel way. However, vertical scaling methods are restricted by the computer upper limits of CPUs and suffer from the cost of a super-power computer. In this case, distributed computing-based BD analytic platforms become the mainstream and Apache Hadoop [46] is one of the most popular platforms. The components of a Hadoop stack and its development history are shown in Fig. 5.

To accommodate the incremental volumes of monitoring data gathered from the IoT, a BD analytic platform is essential for an IoT-SPM ecosystem. Moreover, it should be designed to be able to support IoT architecture implementation, which is outfitted with different elements, such as IoT sensor data ingestion, integration, management, analytics, and so on. These elements constitute a complete process to realize the IoT-SPM ecosystem.

1) *For Data Integration:* One of the most appealing factors of IoT is its ability to offer remote control and visibility over enterprise-wide processes, which is achieved through the fine-grained monitoring of various business units and processes. Data integration is necessary for enterprises and decision makers to collate the data gathered from different sources and to gain a complete picture of what is happening throughout their organization and its environment. For data integration, we could use tools, such as Sqoop, Flume, Kafka, and so on.
2) *For Data Storage:* Through the growing IoT sensor network, sensors constantly gather valuable data regarding different aspects of the business, in different forms and in unprecedentedly large volumes. To transfer and process the IoT data quickly throughout the BD framework-supported IoT architecture, effective data storage needs to be considered appropriately using HDFS, HBase, Kudu, etc.
3) *For Data Analytics:* This is the ultimate goal of the previous steps, such as data ingestion and integration. The data analytics layer enables integrated raw data to be converted into actionable business insights. Decision makers can monitor data and make proactive data-driven decisions via live dashboards under the support of MapReduce or Spark.

4) Computing Resource Distribution: In an IoT architecture, BD platforms and cloud computing are often used together and integrated as two complementary technologies. Taking advantage of its high computing and storage capability, cloud has been used as an efficient tool to process data. However, the cloud computing is based on a centralized paradigm. All the data analysis tasks need to be performed in the cloud, and all the involved data have to be transmitted to the centralized cloud to be processed. Since data transmission depends on network connectivity, cloud computing shifts much of the burden of data processing to an Internet connection. Although the data processing speed can be improved significantly with the support of cloud computing, cloud computing faces several challenges. Low network bandwidth becomes a bottleneck for transmitting such a huge amount of data to the cloud. The long distance between the cloud and the end devices can be an issue for latency-sensitive applications. Some IoT systems might require an immediate response and mobility support, such as fault detection monitoring. In addition, whenever the local data is sent to a third-party cloud service provider, the data owner should consider the security issue and make sure the provider will keep the information completely safe, and sensitive data leakage is not allowed. In other words, these challenges related to network bandwidth, latency, reliability, and security, are the main factors in developing a more reliable solution in terms of cloud infrastructure.

Motivated by the challenges posed when dealing with complex large-scale IoT applications, the cloud computing

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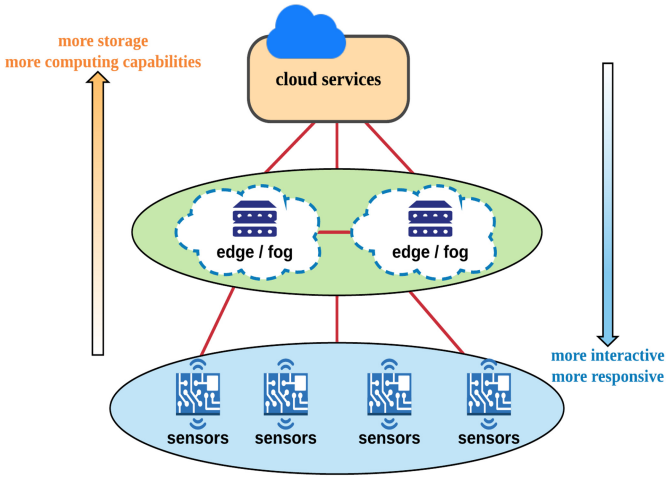


Fig. 6. Example of the new computing paradigm.

infrastructure is evolving to new computing models, with fog and edge computing being the two most common. Fig. 6 demonstrates a general architecture of such a new type of computing paradigm. The work in [47] reports the future directions of next-generation cloud computing paradigms, and points out the traditional two-tier applications will be replaced by new multitier cloud architectures.

5) Machine Learning-Driven Data Analysis: The aim behind these series of steps from data ingestion to setting up BD processing platforms, is to prepare for data analysis using machine learning tools. However, most machine learning methods are task specific and cannot be used for general purposes. To accommodate the new emerging IoT systems, traditional machine learning methods might need huge transitions in terms of the following three aspects.

1) *New Type of Data Generated From IoT Sensors:* The IoT data not only inherit the 5Vs from BD, but also have their own characteristics. For example, industrial companies commonly use vibration, speed, and temperature to track the operations of a piece of manufacturing equipment. Engineers can monitor the health state of a machine across the whole production line in a real-time manner, and arrange proactive maintenance. In this case, the input data should be viewed as a dynamic time series instead of static data and new concepts such as anomaly are introduced. Different data patterns might have their own specific meanings and represent various valuable information in the context of IoT. So, more data mining techniques based on the modified machine learning methods are needed to manage the bulky data particularly created by IoT devices.
2) *New Tasks Released From IoT Applications:* New methods should be developed based on the traditional ones, and especially designed for various applications and requirements newly emerging from IoT systems. With the growing utilization of IoT devices and an increasing number being connected to the Internet, the survivability and self-organization of networks has become an extremely challenging issue and the privacy of the collected data is highly critical as well. In addition, implementing data mining tasks in a real-time manner plays a significant role in real manufacturing and other various fields, such as IIoT, transportation, and healthcare.
3) *New Computing Paradigms Required by Real-Time Analytics:* In general, for a large-scale IoT system, the data mining technologies should be restructured, otherwise they can only be utilized for small-scale laboratory simulation data. Furthermore, there is a growing demand for real-time processing and an immediate response to the end users. Several methodologies have been proposed in the literature to realize data analytics in a real-time manner, such as distributed computing based on the cloud [48] and mobile edge computing [49]. Although the cloud provides strong computing power for fast data processing, implementing the models in a distributed paradigm can make full use of cloud computing. Instead of putting all the computation tasks into one node, distributed computing divides one job into several parts and assigns them into different cloud nodes for parallel computing. Distributed computing still remains inaccessible to a large number of machine learning methods, in spite of many open-source platforms, such as Apache Hadoop and Apache Spark. For example, Spark has offered some programming guidelines in MLlib for data scientists' easy access, such as support vector machine (SVM) classification, logistic regression, Naïve Bayes, *k*-means, principal component analysis (PCA), and so on. There are still many machine learning methods which need more research on how to arrange parallel computing in a distributed environment.

III. IoT-SPM Ecosystem: Current Progress

This section starts by reporting the premier research advances made in IoT domain and related to enabling technologies and components, which are the main trigger and foundation of an IoT-SPM as a promising solution for smart maintenance in manufacturing. In what follows, these advances result in the prosperity in IoT-SPM ecosystem proposal and adoption.

A. IoT Architecture

With the emergence of IoT technologies, the concept of Industry 4.0 and IoT-related ecosystems have attracted the attention of researchers. The increasing number of IoT architectures [see Fig. 7 (top)] has not yet converged to a reference model and there is currently no widespread acceptance of any one particular framework [57], [58], [59], [60]. From the pool of proposed architectures, it can be seen that the basic model is a three-layer architecture [50], which generally consists of the device layer, network layer, and application layer. With the rapid development of IoT, many auxiliary functions are required and emerged, and the rudimentary three-layer architecture cannot fulfill the new demands thus is refined into a more concrete five-layer architecture [32], [51], [52], which includes sensing layer, network layer, middleware layer, application layer, and business layer. The network layer still does

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TABLE I
EXAMPLES OF GENERAL IoT ARCHITECTURES

IoT architecture	references						
i-th layer	[50]	[32], [51]	[52]	[53], [54]	[55]	[56]	
5		business layer	application layer			application service	
4		application layer	middleware layer		interface layer	data integration	
3	application layer	network layer	blockchain layer	cloud application layer	service layer	network transmission	
2	network layer	middleware layer	network layer	edge layer	middleware layer	communication auxiliary	
1	device layer	sensing layer	physical layer	device layer	perception layer	sensing/execution	

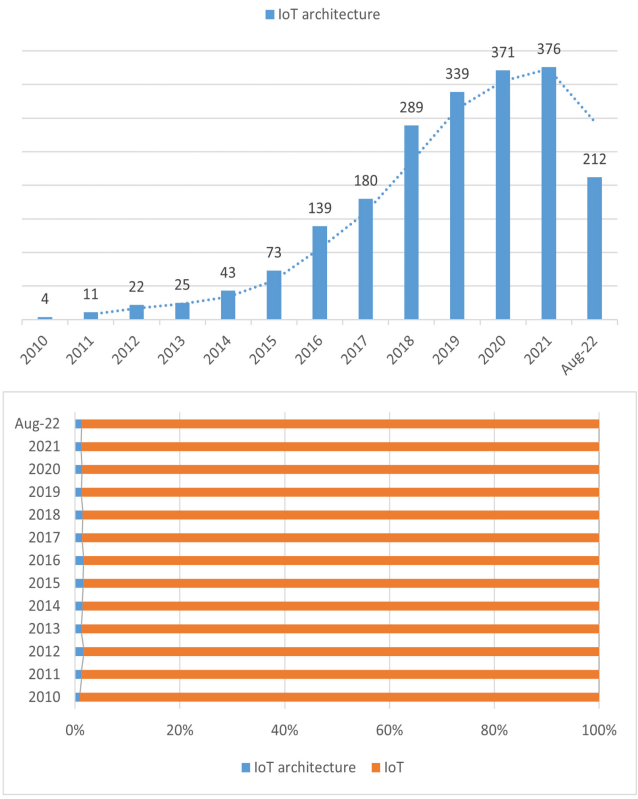


Fig. 7. Publication per year for IoT architecture (top) and percentage for IoT architecture and IoT (bottom). Data retrieved from Scopus (August 1th, 2022) by using these commands when querying this database—IoT architecture: ALL("IoT∗ architecture∗") OR ALL("Internet of Thing∗ architecture∗") AND PUBYEAR = 20XX; IoT: ALL("IoT∗") OR ALL("Internet of Thing∗") AND PUBYEAR = 20XX, in which XX represents the actual year.

the data transmission task. A middleware layer is expanded from network layer and sits between network and application layer to handle the compatibility problems caused by heterogeneous devices with different operating systems and data formats. Recently, Lao et al. [52] proposed a general IoT-blockchain architecture to integrate the blockchain technology into IoT systems. Built on the basic reference model, more particular architectures are explored as application-specific solutions to be applied in specific domains. The work in [22] presents a detailed picture of the state-of-the-art IoT architectures in various domains based on 129 research papers. Based on the general three-layer architecture, the work in [21] gave a deep dive into the exploration of improving IoT architectures regarding networking and communication aspects. A more recent review of IoT architecture was given in the work [38]. Table I lists some typical examples of existing IoT architectures.

B. IoT Components

Whereas some researchers focus on the advances made in relation to the entire IoT architecture, it only account for a small portion of vast IoT-related studies [see Fig. 7 (bottom)]. There are also many other researches focusing on a specific part of the IoT architecture-based ecosystem with an emphasis on enabling technologies. Across an IoT-based ecosystem from physical to cyber space, the integrated design and development of both hardware deployment and software applications are equally essential and beneficial to the IoT-SPM ecosystem. He et al. [61] overviewed the recent advances and the current progress in hardware and software from sensing, networking to analysis in IIoT. Six elements are surveyed to gain a better insight into how the functionality of the IoT is delivered, namely, identification, sensing, communication, computation, services, and semantics [50]. A nonexhaustive list of recent works on specific IoT components is showing in Table II. It is of note that the exponential growth of IoT-related studies in research community makes it impossible to provide a full list of publications. Our intention is to bring it to public consciousness that IoT ecosystems are established by diversity of components driven by enabling technologies from multiple disciplinary and the required components are not just limited to the ones showed in Table II.

The work in [40] reviews the recent trends in automation technologies and the impact they may have on industrial communication in the era of IoT and Industry 4.0. The potential of fifth-generation (5G) and Ethernet time-sensitive networking (TSN) for industrial automation applications is highlighted. In the current stage, WSNs are the most important components of IoT, and energy efficiency is a critical issue especially for sensor nodes which run on batteries and are generally difficult to recharge once deployed. This also has triggered numerous research motivations into wireless and battery-free sensing technologies over the past few years. Burg et al. [42] focused exclusively on the myriad of wireless communication systems with standards, protocols, and security issues. Xu et al. [62] generated a connected and efficient network topology for data collection in large-scale WSNs using a joint clustering and routing design. The work in [63] provides a novel data collection approach named RMER (reliability and multipath encounter routing) to design energy-efficient and reliable sensor communications protocols for IoT. By deploying fewer monitor nodes in hotspot areas and more nodes in nonhotspot areas, multipath routes are converged together outside nonhotspot regions such that the aggregated data can be sent to the sink in an energy-efficient way. In [64], a series of trust evaluation criteria is utilized to evaluate the sensor nodes. Based on the trust values assigned to each node, a fog

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TABLE II
NONEXHAUSTIVE LIST OF IOT-RELATED RESEARCH STUDIES AND CATEGORIES OF IOT COMPONENTS THEY ARE FOCUSING ON

references	IoT components						
	sensing	networking & communication	data security	data collection	data storage	resource allocation	analysis
Wollschlaeger et.al. [40]	✔						
Burg et.al. [42]		✔	✔				
Xu et.al. [62]				✔			
Dong et.al. [63]				✔			
Wang et.al. [64]				✔			
Duan et.al. [65]		✔					
Wan et.al. [66]		✔					
Jiang et.al. [25]					✔		
Chervyakov et.al. [67]					✔		
Cai et.al. [26]					✔		
Yang et.al. [28]			✔				
Wang et.al. [68]			✔	✔			
Luo et.al. [69]			✔	✔			
Tao et.al. [70]			✔	✔			
Li et.al. [71]			✔	✔			
Yang et.al. [28]			✔				
He et. al. [61]	✔	✔					✔
Landaluce et al. [72]	✔	✔					
Meng et al. [73]	✔	✔					
Cui et al. [74]	✔						
Lin et.al. [75]						✔	
Li et al. [76]						✔	
Zhang et al. [77]						✔	
Xiong et al. [78]						✔	
Li et al. [79]						✔	
Li et al. [80]						✔	
Carvalho et al. [81]							✔
Zhang et al. [82]							✔
Mahdavinejad et al. [83]							✔
Nnamuduri et al. [84]							✔
Serradilla et al. [85]							✔

node-based data collection path is designed to guarantee the reliability of the network nodes and the credibility of the collected data. The work in [23] reviews a variety of economic and pricing models to help develop adaptive algorithms and protocols for WSNs.

The works in [79] and [80] focus only on resource distribution in the case of convolutional neural networks (CNNs) by designing an offloading strategy to offload the computational burden from the central cloud to the fog nodes. But the adopted CNN requires point-to-point labels as the ground truth to train the model. The data set used in the experiment was manually collected and hundreds of images are not enough to be considered BD. More importantly, its real-time processing ability is not verified, which should be the top priority when deploying it in an industrial environment. In contrast, papers, such as [65] and [66] only focus on efficient data exchange and network transmission, demonstrating that the software-defined network (SDN) is an impactful approach for industry.

Jiang et al. [25] proposed a framework especially for data storage which combines multiple databases and Hadoop to store and manage diverse types of data including both structured and unstructured data. The work in [67] proposes a configurable reliable distributed data storage system which not only increases safety and reliability, it also reduces the overheads of data storage and the processing of encrypted data. The cloud-computing-integrated IoT is analyzed and discussed in [86]. Yang et al. [28] thoroughly explored the security and privacy issues related to IoT in four steps. The limitations of IoT devices and their solutions are first presented, followed by the classification of IoT attacks. Next, the mechanisms and architectures for authentication and access control are investigated, after which the security issues are analyzed and discussed. The work in [83] assesses the various machine learning methods when applied to IoT data to extract higher-level information. The work in [24] provides a comprehensive review on the integration of IoT and existing network systems, including the Internet, cloud computing, BD, social, and industrial networks. There are also some IoT-driven frameworks that integrate several IoT components, developed for various applications, such as health monitoring [31], smart manufacturing [48], [87], smart cities [88], smart agriculture [89], [90], smart grids [91], smart transportation [92], smart farming [93], [94], and so on.

In recent years, the number of research works related to IoT has increased dramatically. It is not easy to cover all the previous studies in this section. We present several papers in relation to reviews on IoT and researchers who are interested can refer to the details. Yang et al. [95] presented an overview of the IoT, starting with a discussion on the key technologies that integrate things to IoT ecosystems, including RFID, WSNs, and mobile computing. Then, IoT data protocols that are used for interconnection between these different things are also reviewed. The work in [95] also investigates IoT platforms and IoT technologies. The work in [96] summarizes and analyses the requirements of smart manufacturing systems from four perspectives, namely, objective, functional, technological, and business requirements. Wang et al. [97] reviewed the blockchain concept and features in a systematic way and summarize how the blockchain is applied to IoT applications using its security tools and technologies. Mistry et al. [98]

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presented a comprehensive review on blockchain-based 5G-enabled IoT and discuss the research challenges related to their integration for industrial automation. Fahim and Sillitti [99] presented a systematic literature review of studies on anomaly detection, analysis, and prediction techniques in the IoT environment. Dehkordi et al. [100] conducted a survey on data aggregation in IoT sensor networks. The work in [101] provides a detailed review on IoT security, including the related challenges, threat sources to IoT applications, and existing technologies. The work in [102] provides a review of the edge computing technology and its reference architectures.

These papers (Table II) address or investigate one or several issues emanating from the diverse challenges in IoT systems. They account for a large portion of IoT system-related papers, while a few articles address all the challenges in one solution and present a complete IoT system. The next section reviews and compares the recently proposed IoT systems and delves into the details to conduct an evaluation qualitatively.

C. IoT-SPM Ecosystem

The emergence of various IoT technologies has led to Industry 4.0 and many research efforts have been devoted to the IoT-SPM ecosystem. Table III summarizes the proposed ecosystem and used technologies. The work in [103] proposes a smart prognostics and health management (PHM) IoT approach for smart maintenance. The proposed PHM architecture involves data acquisition and processing, fault detection and diagnostics, and fault prognostics and decision support.

Wan et al. [87] proposed and implement a cloud-assisted architecture to process BD in the cloud for active maintenance in manufacturing environments. The proposed solution also includes the collection of manufacturing BD using OPC UA and network communication. BD analysis is performed in the cloud and the cloud layer architecture is designed to realize both the mechanism of real-time active maintenance (MRAM) and the mechanism of offline prediction and analysis (MOPA) using neural networks. The experiments show the advantage of this new type of solution compared with the traditional maintenance mode in terms of improving the tool working time. This research is an initial work in the development of an IoT-based ecosystem for smart maintenance and is a primary form of IoT system. In this work, data collection and transmission and cloud computing have already been integrated into manufacturing maintenance, but most efforts are concentrated on cloud processing. It did not undertake any in-depth research on data integration, storage, or communication.

Lee [104] provided an IoT-embedded cloud control architecture to control multisite and multiproduct manufacturing plants and processes. The deployed IoT devices are able to detect system status and transmit signals to a central place where deep learning-based analytics can be utilized to process the data. The proposed architecture consists of product site, IoT beacon, network architecture, communication protocol, and cloud control server. The work in [105] presents a conceptual and initial structure to exploit IoT for the smart maintenance of medical equipment. Basic hardware and software requirements for its implementation are also discussed. Niyonambaza et al. [106] proposed a predictive maintenance structure using IoT to predict the early failure of mechanical equipment in Rwandan hospitals before failure occurs. The structure design consists of a simple real-time data collector prototype and predictive data analytics for equipment health status classification. Moens et al. [115] adopted a scalable three-tier IIoT architecture for smart machine maintenance in IIoT applications. The first tier involves the edge nodes, including the sensors, actuators, and control systems connected to the physical machines, which can be connected to the edge gateway with the access network. The first tier is then followed by an IoT middleware cloud platform for data ingestion and persistence and a dynamic dashboard application for monitoring and visualization.

Civerchia et al. [107] presented an advanced IIoT solution for predictive maintenance applications, aiming to enable a pervasive monitoring of industrial machinery through battery-powered IoT sensing devices. The proposed system is based on four components: 1) sensor devices; 2) gateway; 3) remote control and service room; and 4) an open platform communications server. Zhong et al. [108] introduced an IoT-enabled real-time machine status monitoring platform, which is divided into several layers: smart machines outfitted with different sensors to capture machine data, smart connection for data communication, cloud-based services developed to make full use of the captured data for different decision makings, and a smart view layer designed as an end-user interface which provides augmented reality-based visualization services. Li and Kara [109] proposed the integration of both WSN and IoT into one system architecture for environment monitoring in manufacturing. The proposed generic architecture is split into four layers: 1) data acquisition; 2) WSN interfaces with a microcontroller to deliver the data; 3) cloud services; and 4) data visualization through a Web browser or a mobile application.

With the emergence and porosity of Industry 4.0 and the IoT, the increased amount of monitoring data motivates researchers to investigate more innovative ways to manage such volumes of data. At this stage, a number of technological requirements arise, including the efficient management of thousands of messages per second without leading to network conjunction, the fast processing capability for a real-time purpose, storage with high throughput, the ability of quick querying, and so on. Driven by these challenges, Canizo et al. [110] presented a large-scale platform for the CPS real-time monitoring of industrial machines in a real work environment. To process the BD that are continuously generated from many industrial machines, a BD architecture is proposed to manage the large volume of data from the CPS. Data acquisition, data processing, data persistence, and data serving are introduced into the architecture as components. The platform can be divided into three blocks: 1) local data acquisition; 2) a cloud platform; and 3) a front end. Thus, this platform is still a centralized structure and all the computing resources are concentrated in the cloud. This article does not involve any specific data analysis algorithms, and the experiment only tests the performance and scalability of the real-time processing side. In comparison to the works discussed previously, the proposed system for smart maintenance in the work in [110] is becoming more complete.

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TABLE III
VARIOUS IOT-SPM ECOSYSTEM AND USED TECHNOLOGIES. BLANK CELLS INDICATE NO CORRESPONDING TECHNOLOGY IS EMPLOYED OR REPORTED

References	sensor data	communication	big data platforms	computing strategy	ML-based methods	use cases
Wan et al. [87]	alarms, device logs & status	Zigbee or WiFi	XenServer & Hadoop	cloud	data processing	machine tool
Lee et al. [104]	gap width, etc.	IEEE 802.11ac(Wi-Fi)	Intel's IoT analytics	cloud	deep belief network	HID cable
Ranjbar et al. [105]	test result data	WEC/Wi-Fi		cloud		medical equipment
Niyonambaza et al. [106]	temperature, etc.	GSM			LSTM	Mechanical Equipment (pumps)
Civerchia et al. [107]	temperature & vibration	IEEE802.15.4, 6LoWPAN CoAP protocols				Industrial machinery
Zhong et al. [108]	RFID sensor data	wire & wireless		cloud		manufacturing machine
Li et al. [109]	temperature	wireless	Data.Sparkfun®	cloud		manufacturing environment monitoring
Canizo et al. [110]	temperature, pressure, inductive or flow meter sensors, etc.		Apache Spark	cloud	anomaly detection algorithm	press machine
Yu et al. [48]	temperature, pressure, etc.	SFTP, VPN gateway	Apache Spark	cloud	DPCA	turbine syngas compressor
Chen et al. [111]	equipment execution data	ad hoc network with DDS protocol	Hadoop	edge		candy product production line
Wu et al. [112]	force, rotational speed, temperature, vibration, acoustic emission, and torque sensors, etc. ($>$ 50 sensors)	ZigBee	Apache Spark	fog	random forest	pumps and CNC machine
yu et al. [113]	temperature, pressure, etc.	SFTP, VPN gateway	Apache Spark	edge	autoencoder	syngas compressor
Ayvaz et al. [114]	motion, speed, weight, temperature, electrical current, vacuum and air pressure, etc. (101 features)	gateway devices			automatic selection from random forest, XGBoost, Gradient boosting, Adaboost, MLP, SVR	production assembly lines
Hosamo et al. [17]	BIM model's data & sensor data& FM system's data	network with BACnet protocol			expert rules based on APAR /ANN, SVM, decision tree	air handling unit

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Yu et al. [48] proposed a BD ecosystem that could be easily implemented in an Industrial 4.0 system for fault detection and diagnosis in predictive maintenance with real industrial BD gathered from large-scale global manufacturing plants. The proposed system provides a complete solution from data ingestion to data visualization in a real-time environment, where various technologies, such as data lake, NoSQL database, Apache Spark, Apache Drill, Apache Hive, and OPC Collector are utilized. For BD analytics, MapReduce-based distributed PCA (DPCA) is introduced for immediate processing, ease of implementation, and extension to a wide range of application. The experimentation based on a real case study shows the system is able to provide an alarm warning several days before the fault happens. With the advances of information technologies in all aspects of industry, large volumes, and more modalities of data are available and more advanced data mining algorithms are considered. However, in the premise of accuracy, requirements for robustness and real-time processing are the most demanding and critical parts of the entire system and play more important roles in the real industrial environment. Algorithms such as PCA which may not be that advanced but with the advantages regarding efficiency might be the most suitable ones. In summary, the proposed ecosystem in [48] is capable of handling BD and partial labeling issues, while the data preprocessing is undertaken completely manually based on expert knowledge. All the computing tasks are assigned in the cloud center, and no resource distribution optimization is involved.

The work in [111] proposes an architecture of edge computing for IoT-based manufacturing. The incorporation of edge computing allows the IoT system to gain the ability of mobility, geographic independence, and low latency. Edge computing is introduced in the system in a schematic theory-like manner and the discussion is very conceptual in scope. Wu et al. [112] presented a system architecture based on fog computing and IIoT to collect real-time streaming data and monitor machine health conditions and processes. The system architecture includes data acquisition, data communication under the standard communication protocols and predictive analytics and integrates these tasks with the private and public cloud. Training data sets are streamed to a remote public cloud to build diagnostic and prognostic models, while real-time data sets are collected and streamed into a local private edge cloud for online diagnosis and prognosis. The work in [112] also only provides ideas about introducing fog computing into the IoT system and even implementation details such as software-related discussions are missing. Furthermore, no context on data issues is given in this article.

IV. IoT-SPM Ecosystem: Data Quality

A. Gap Identification

In light of the aforementioned background introduction and discussions on IoT-based ecosystems for smart maintenance, a large body of research has been conducted in terms of developing IoT architecture and optimizing each component to meet the new requirements of Industry 4.0 systems. Huge efforts have been made to realize the IoT concept in the real world, including data collection, data transmission, data storage, data management, and data analytics. A lot of attention has been given to the architecture, ecosystem, and infrastructure and many have been set up in real manufacturing. IoT technologies have already been developed to a very mature stage. In addition, data aspects are also very important since smart analytics is not realized by the infrastructure, it is actually driven by the data. The quality of the system or infrastructure completely depends on the data. We can have a good architecture or system, but we will not achieve a good outcome from smart analytics if we do not have proper data. This motivates us to present this section especially focuses on data quality.

Most existing efforts [81], [82] have gone into predictive maintenance applications for IoT-based smart maintenance, which focus on machine learning algorithms and particularly data-driven methods. High-quality data can only achieve high-quality data mining results. In the modern IoT paradigm, data quality has become a very important aspect that influences the overall performance of any system. The work in [61] points out that one of the characteristics of IIoT data is the low quality of raw data and data cleansing is indispensable for advanced data analysis algorithms. As discussed in [116], one of the major challenges that affect IoT is data quality and uncertainty, because as the data volume increases, noise, inconsistency, and redundancy increases within the data and causes serious issues for IoT technologies. The work in [117] also identifies data quality as one of the key IoT challenges.

Many research projects utilize machine learning, spending 80% of project time on data preprocessing and 20% on model making. Data quality can directly affect the accuracy of data analytic models and poor-quality data can mislead the decision-making process [118], [119]. Liu et al. stated in [120] that data are crucial to support decision making. If data have low veracity, decisions are not likely to be sound. Without addressing data quality issues, it is useless collecting and managing IoT data by developing an IoT architecture since the data is unreliable. The study in [119] proposes that IoT capabilities can help create significant value in business processes if supported by good-quality data and better data quality is needed to unlock the value of IoT capabilities. Analysts have estimated that firms lose 8% to 12% of their revenue due to poor-quality data.

However, only a few researchers have studied the IoT data quality issue, which is the most important issue in the development of an IoT-based ecosystem in real industrial manufacturing practice. Especially in IoT manufacturing systems, real-world data collection is normally handled by industry engineers, and there are few connections between industry and academia, in part because of commercial secrecy [1]. Most academic research focuses on improving modeling techniques based on public data sets, which are always quite clean, small-scale, outdated, resource limited, and impractical. Due to the lack of opportunity to access the field data generated from real industry, very few researchers consider topics, such as optimizing data and possible data quality issues. So, most existing modeling techniques reported in the published papers cannot always be directly patched together and applied to real IoT manufacturing plants. This also explains why most smart maintenance systems that have been deployed in real production are only found in large companies, because it is necessary

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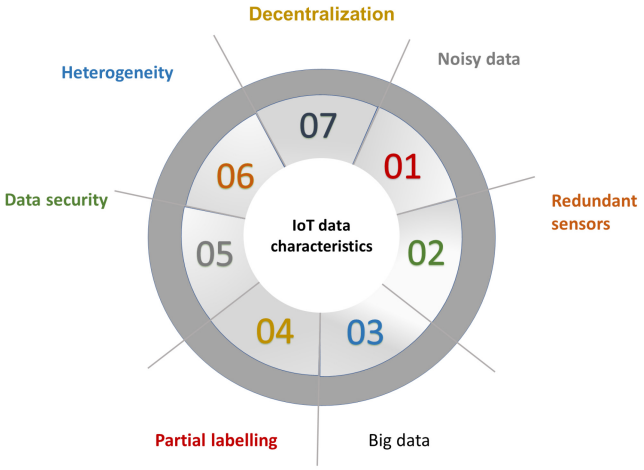


Fig. 8. IoT data characteristics.

but difficult and expensive to assemble industry experts and scientists in academia. Therefore, there needs to be a clear understanding of data quality for real industrial applications, the challenges associated with data quality, and a way to evaluate and ensure that data quality is maintained for IoT applications. Looking at the IoT ecosystem as a complete picture, we identified that in most IoT-enabled frameworks, there is a gap between IoT-collected data and data analytics using machine learning methods, which is the data preprocessing that is responsible for converting the raw data into data of a quality that is analysis ready.

B. IoT Data Characteristics

Data represent a valuable asset in the IoT paradigm as a source for extracting insights and a means of communication [121]. Over the past few decades, manufacturing data have evolved from the handicraft age to the machine age and information age. With the pervasion of IoT technologies in the industrial area, a huge amount of IoT sensor data is being generated every day and everywhere, so manual feature engineering is no longer feasible. Such a transition fosters a new kind of data lifecycle, where smart things are the main actors. Things in IoT are able to communicate and collaborate with each other through IoT data to autonomously provide intelligent services. The data give insights about a given entity, machine, or phenomenon, and these insights can help various applications with smart outputs in a ubiquitous manner. Generally, as shown in Fig. 8, IoT data have the following characteristics.

1) *Noisy Data:* In general, each IoT paradigm has an IoT architecture as the technical support to provide end-to-end services. As there is a long process to go through before IoT data is stored in an analysis-ready format, a small error or failure at any step will endanger the quality of data, and the generated IoT data will be inherently erroneous and noisy. The work in [121] gives more details about the possible factors that might affect IoT data quality.
2) *Redundant Sensors:* To achieve seamless monitoring, the IoT system installs high-density sensors at all possible points to capture as many features as possible to ensure the physical objects are fully described and well digitalized. It is not necessary nor is it practical to feed all the variables collected by the sensor into the analytic model. Many sensors are situated geographically close to each other thus might highly correlated to each other numerically. Some sensors also deliver contradictory features.
3) *BD:* People usually relate IoT data with BD, which refers to large amounts of multisource, heterogeneous data generated throughout the product lifecycle. The collected IoT data is generally characterized by 5Vs [122], which is volume (i.e., a huge amount of data), velocity (i.e., the high speed of data generation), variety (i.e., data flows in different formats and from heterogeneous sources), veracity (i.e., data is associated with inconsistency, bias, uncertainty, and noise), and value (i.e., users need to extract useful information and explore the hidden values from BD). More details can be found in [123], where all the data issues are related to these 5V features.
4) *Partial Labeling:* Due to the high frequency of IoT BD, it is beyond our ability to obtain fine-grained point-to-point labeling results. Thus, the demand for unsupervised or weakly supervised analytics might be very high and will be the top priority in certain specific situations.
5) *Data Security:* The IoT is the network of physical objects or things that are embedded with sensors, software, and other technologies to connect and exchange data with other devices and systems over the Internet, hence cybersecurity is a critical factor that needs to be considered carefully.
6) *Heterogeneity and Decentralization of Data:* IoT devices use diverse positioning technologies with not unified standards and protocols, potentially resulting in incompatible formats, resolutions, and semantics. This one also corresponds with "variety" in 5Vs of BD characteristics.

C. IoT Data Quality Issues

With particular attention on data quality and data difficulties in the context of IoT manufacturing, the possible issues that might be encountered when dealing with real industry cases are illustrated. In this section, we focus on the quality of IoT monitoring data, and the difficulties when processing them.

1) Determine Which Types of Data to Collect: There is much to be done to realize the potential of digital twins [1] by utilizing IoT technologies and similarly for IoT-SPM. Since the data is the base of the whole system, the first step is to decide what types of data to collect, but it is not always obvious and is usually task specific. For example, to monitor machine health, we might require vibration, speed, power, temperature, pressure, together with environmental conditions, such as wind direction, humidity, and so on. In this case, predictive maintenance can be performed based on equipment data collected from smart manufacturing by IIoT technologies, which includes operating conditions, surrounding environments, and the maintenance history of production equipment [48]. To make production planning and shop floor

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scheduling more intelligent, customer orders, supply chain data, production capacities, and inventory data can be collected and analyzed. In this case, academic research might not be practical if it is isolated from industry since the modeling in research highly depends on the data and needs to be carried out based on the real data from industry to make it feasible and practical in industry.

2) Various Standards and Ownership When Collecting Data: It can be difficult to aggregate data from thousands of sensors. Data can be spread among many owners and be held in various formats, and scattered ownership of data is a great barrier to data aggregation [124], [125]. Disparate data types are difficult to merge, such as vibrations recorded as lengths of time or frequencies, images saved in different scales, time-series data sampled at different rates, and so on. Fortunately, these issues can be initially addressed by collecting data under a unified standard. For example, in one smart factory, IoT-related sensor data can only be collected under certain rules, in which case it would be easier for data aggregation. Unlike creating a big complex system across a city or country, the data ownership issues do not exist in a factory which is held by one organization. However, to achieve the ultimate goal of a future smart universe which is envisioned to connect an increasing number of things into one system, these issues have to be addressed in an appropriate way and relevant standards and protocols need to be designed.

3) Noise Detection in the Presence of Anomalies: One of the characteristics of IIoT data is the low quality of raw data [61]. Unlike the small-scale data collected from laboratories, IIoT data from real industrial settings are always associated with noise, and noise removal is necessary to improve the performance of further data mining algorithms. However, IIoT also introduces a new type of data, which is known as anomalies. Noise and anomalies both belong to outliers, referring to the points which deviate from normal points. They are related but distinct in concept and have totally different meanings during the manufacturing process in the context of IIoT. In manufacturing monitoring data, an outlier may have two interpretations, that is, it can either be noise or a significant anomaly [126]. Noise may appear as inaccurate data resulting from sensor errors, while anomalies can be items of interest caused by abnormal equipment conditions. Therefore, anomalies are meaningful to a data analyst as potential indicators of significant events, while noise acts as a hindrance and might lead to model misspecification and incorrect analytics. Based on these considerations, new challenges are introduced to noise detection for IIoT-based intelligent monitoring systems, which is to precisely extract noise in the presence of significant anomalies and preserve the anomalies at the same time. To demonstrate this, Fig. 9 shows a visual comparison of noise and anomalies in field data collected from a real manufacturing plant.

4) Missing Value Imputation: In addition to noise, missing data must also be handled accurately and efficiently to improve data quality [127]. Deriving the appropriate values to impute the missing points is important, since many data analysis methods require serially complete observations. All decision-making processes make use of the available information as

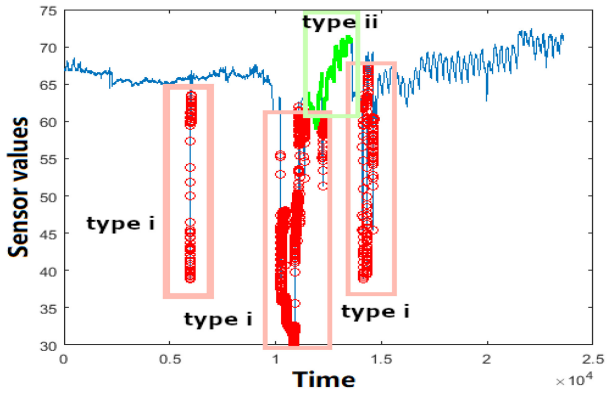


Fig. 9. Different types of outliers. Points marked in red are type i outliers known as noise. Points in green are type ii outliers known as anomalies.

model inputs to generate analysis results, but the presence of missing values in the available information may lead to the breakdown of an analytics model. Many practical applications and statistical methodologies, such as spectral analysis and down-scaling, exhibit no tolerance to missing values. Data structure and patterns are broken due to missing values, in which case decisions made based on such data are biased and unreliable. Hence, it is essential to consider and handle the missing value issues in IIoT.

5) IoT Sensor Selection: The optimal number and placement of sensors need to be determined to make sure all the valuable information is collected in an appropriate way. If only a few sensors are installed to monitor machines, the prediction results will be inaccurate based on limited data. In contrast, too many sensor inputs will distort and obscure the outputs [128]. Features which do not provide useful information for the outputs might degrade the modeling performance. Therefore, sensor selection, which identifies only the useful sensors that will be used in further data analysis, is a necessary step to obtain high-quality IoT sensor data.

Especially in a plant-wide monitoring system, a large number of machines need to be monitored, but each sensor is only responsible for one or several specific pieces of equipment. If a sensor which is not related to this piece of equipment is taken into consideration in the monitoring model, it might negatively affect the fault detection results. Thus, one general model will be not feasible for detecting all types of faults for every machine. Motivated by this issue, multiblock-based techniques have been introduced and developed to detect different types of faults with separate models. With regard to one certain fault, only the responsible sensors are identified and selected as the inputs to build a specialized model. In this way, all the sensors are divided into several blocks to monitor different parts of the system independently, and different models can be deployed separately and distributed into several edge nodes [129].

6) New Issues in Result Analysis: When a fault is detected by the monitoring model, contribution analysis of all the incorporated variables is required to track which sensors have contributed to exceeding the alarm threshold and identify which part of the system or which machine has broken down. By locating the sensors responsible for the abnormal behavior,

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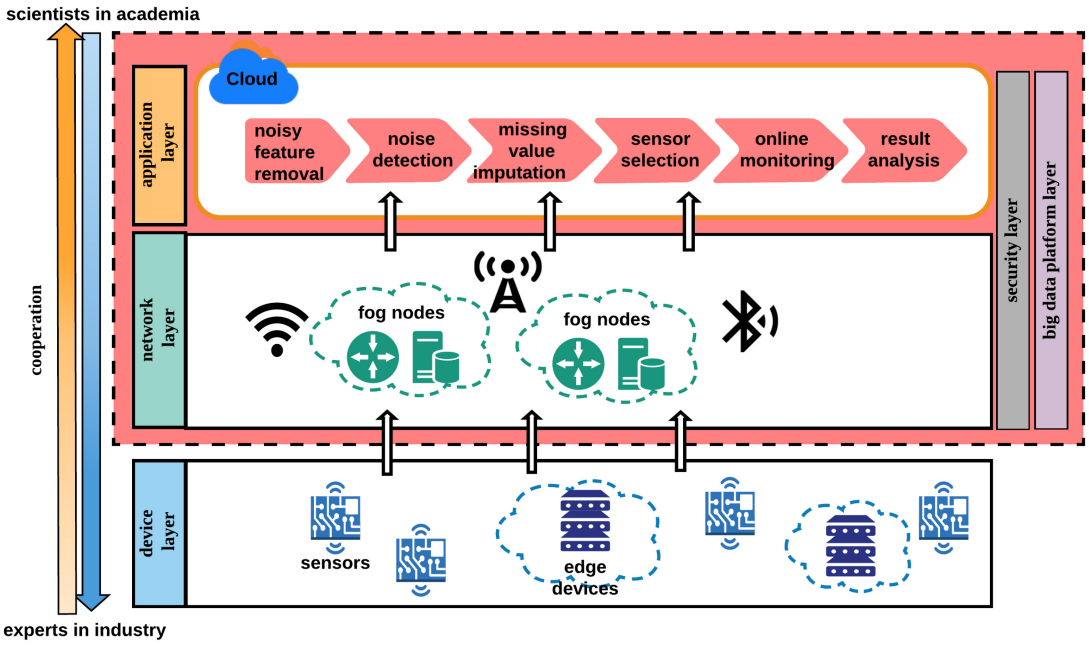


Fig. 10. Proposed reference IoT-SPM ecosystem.

engineers can easily identify which machine is in trouble and take action in a timely manner to prevent further damage to the whole production system. However, a new issue arises, namely, what if the malfunctions occur in the sensors not in the monitored machine, in which case a fault with the sensor will be mistaken for a machine fault. So, distinguishing machine faults from sensor faults is important to help make smarter decisions and it is also necessary when generating a comprehensive and intelligent report. One of the possible solutions is to use contribution analysis results to classify whether a fault stems from a machine or a sensor. In a smart factory, it is common that every operation unit in the production system has several sensors installed to achieve 360° monitoring. When a machine-related fault occurs, multiple sensors are affected, which is reflected in their measurement readings, hence they can be identified as the sensors which contributed to the fault. Otherwise, if only one sensor contributes to the detected fault, and the rest of the sensors behave normally, this means the problem occurred in this sensor.

V. Data-Focused Reference IoT-SPM Ecosystem

Our initial works [48], [130] focused on the design and implementation of an IoT architecture which has been deployed in a real industrial company. When we conducted a deeper study, more issues regarding practical applications were discovered. These findings triggered our further research on noise cleansing [126], [131], missing value imputation [132], and sensor selection [129]. In light of the discussions in previous sections and our previous study, we propose a reference IoT-SPM ecosystem in this article, and this standard ecosystem can be extended to accommodate various requirements in specific scenarios. The schematic architecture of the proposed ecosystem is shown in Fig. 10.

First, sensory devices that are capable of monitoring diverse physical metrics but rely on fewer resources are deemed to be smarter and more practical in IIoT. Then, new forms of networking and communication are expected for efficient data transmission. In the application layer, the proposed data processing workflow can be implemented. For unsupervised learning, every feature is taken into equal consideration when calculating the models. Thus, sensor selection is vitally important to only select the features (inputs) which are related to the detection results (outputs). In contrast, supervised techniques can learn the weights of each feature automatically during the training phase, and in a sense, they contain the exact sensor selection functionality. Broadly, components composing the application layer are summarized as follows. Note that some components can be implemented using new computing paradigms and distributed into edge devices or fog nodes.

1) *Remove Noisy Sensors:* This step preliminarily weeds out the sensors which do not provide accurate information and degrade the model performance. The sensors whose variables have high a percentage of noisy points or missing values are defined as noisy and are removed, since the information from these variables is no longer reliable.
2) *Noise Detection:* Unlike the public data set or the data collected from the laboratory simulation, noise is ubiquitous in the real IIoT sensor data. Removing or repairing noise enables more accurate models to be developed. In the context of IIoT, noise detection methods in the presence of anomalies can be utilized.
3) *Missing Value Imputation:* This component is responsible for imputing the missing points with the most appropriate values and providing a complete data set to be analyzed in the integrity-required applications or stored in the data warehouse.

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4) *Sensor Selection for Different Types of Faults:* To detect different types of faults, separate models build upon different sensor subsets. For this purpose, sensor selection is employed.
5) *Fault Detection:* Based on one certain sensor subset as data input, the corresponding type of fault can be detected effectively. One of the most popular techniques is PCA-based unsupervised fault detection.
6) *Contribution Analysis for Root Cause Detection:* Once a fault is detected, the root sensors that cause the issue are identified through contribution analysis.
7) *Sensor Fault or Machine Fault:* Further analysis on the root sensors can be conducted to distinguish true machine faults from possible sensor faults. According to the analysis results, engineers can arrange the most effective measures to reduce the human and economic cost.

VI. Qualitative Evaluation on IoT Data Quality

To determine the road map of this evaluative study, we first inquire the studies by using the search statements: KEY ("IoT*") OR KEY ("Internet of Thing*") AND KEY ("data quality") in Scopus, generating 78 documents. The exclusion criteria are as follows by fully reviewing the searched studies.

1) The study is not IoT data quality related.
2) The study is written in a different language other than English.
3) The full version of the study is not available.
4) The study is published before 2010.

Through the review process, there are some overview papers. Prior works mostly [133], [134] focus on the data quality dimensions which can also be extended to IoT domains, such as accuracy, timeliness, completeness, usability, trustworthiness, confidence, consistency, redundancy, and so on. This article studies the IoT data quality from the specific tasks level with the purposes to address and improve the quality of data.

Karkouch et al. [121] provided a theoretical basis of data quality in IoT, analyzing the IoT data characteristics and the factors affecting data quality and their impacts. Alwan et al. [135] conducted a review on data quality challenges in CPS, but the scope of discussion is very limited. The work in [136] shows how data quality can be evaluated contextually using two use cases and realizes the importance of improving IoT data quality, but the discussion is still at a high level. Liu et al. [120] discussed the impact of data veracity in decision-making processes, identify the challenges of handling BD from the IoT viewpoint, and recommend several future research directions for veracious data analytics in IoT. The work in [116] points out that IoT is facing challenges and open issues, including hardware, security, privacy, heterogeneity, virtualization, and data analysis challenges. Of these challenges, Ghallab et al. [116] focused more specifically on the outliers' detection challenges of data quality. Ottosen and Kumar [137] applied outlier detection and gap-filling methodologies to low-cost air pollution sensors, since poor data quality delivered by low-cost sensors prevents the widespread adoption of these sensors.

Research into data quality in IoT is still in its infancy and current studies have only just started to notice the importance of bringing the concept of data quality into focus. Most existing discussions are at a high level and in a general sense. There are no papers related to IoT-based smart maintenance-specific cases which detail the techniques to improve data quality in IoT. This article identifies several topics that are categorized into three groups, namely, outlier detection, missing value imputation, and sensor selection. Regarding the above three category tasks, we additionally search the keywords: 1) "IoT" or "Internet of Things" and 2) "outlier detection" or "noise detection" or "feature selection" or "missing value imputation" in Google scholar. In the following, we discuss and evaluate qualitatively the possible solutions to these issues in IoT-specific domains.

A. Outlier Detection

Outlier detection is very popular in industrial applications and many classic methods or tools have been developed and built. Diverse formulations need to be designed to address the specific problem. Especially for IIoT, outlier detection methods are highly specific to the individual use-case [138]. For example, for fault diagnosis-related monitoring data, outliers contain both noise and anomalies [139]. Removing noise removal while preserving anomalies is a new problem and the existing off-the-shelf methods are not suitable in this case. Second, due to the streaming nature of IIoT data, processing algorithms need to be as simple and efficient as possible to meet the timeliness requirement in resource-constrained environments.

Fig. 11 presents a lifecycle of outliers in IoT from sources to detection. Based on the machine learning algorithms used to detect outliers, outlier detection approaches are classified into several types in [140]: clustering-based, classification-based, dimension-reduction-based, and hybrid approaches. In the IoT-specific domain, the sources of outliers can be: 1) noise or error which is resulted from sensor measurement or transmission and 2) events or faults caused by equipment malfunctions. Based on the type of detected outliers in IoT, the related papers can be divided into three types accordingly. Some works are concerned with detecting outliers regardless of which type they are [141], [142], [143], which is exactly what the traditional outlier detection approaches do. With the wide adoption of IoT-SPM, there are a large body of research focusing on anomaly detection which is really a hot topic in IoT research community [144], [145], but very few investigate noisy data cleansing before anomaly detection [146]. Table IV shows a comparative evaluation of outlier detection in IoT.

1) *The Detection of Outliers (All Abnormal Points):* The work in [162] proposed a framework to block outliers that are produced either intentionally (hacked nodes) or unintentionally (fail-dirty nodes), which refers to noise and anomalies in this article. The work in [147] utilizes a k-means clustering-based algorithm to detect anomalous readings. The work in [148] focuses on outliers which are defined as an observed value that is far from the others. In [153], outliers are explained as any abrupt

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TABLE IV
COMPARATIVE TABLE OF OUTLIER DETECTION TECHNOLOGIES FOR DIFFERENT SOURCE TYPES

outlier sources	references	technique	technique category					
			clustering	classification	dimension reduction	hybrid	threshold	other
both	[147] 2015	k-means	✔					
	[148] 2019	the median filter (MF), empirical mode decomposition (EMD), classification and regression tree (CART), etc.				✔		
	[149] 2018	game theory						✔
	[150] 2018	one-class support vector machine-based		✔				
	[151] 2017	PCA-based			✔			
	[152] 2018	density-based spatial clustering of applications with noise (DBSCAN)	✔					
	[153] 2018	Classification and Regression Trees (CART), Random Forest (RF), Gradient Boosting Machine (GBM) and Linear Discriminant Analysis (LDA)				✔		
	[154] 2020	KNN-based					✔	
	[116] 2020	NRDD-DBSCAN	✔					
anomalies	[155] 2016	PCA-based			✔			
	[156] 2020	Autoencoder, LSTM			✔			
	[157] 2017	ARIMA					✔	
	[48] 2019	PCA-based			✔			
	[113] 2022	Autoencoder-based			✔			
	[158] 2016	SVM		✔				
	[159] 2019	CNN-based transfer learning		✔				
	[160] 2019	Spectral Residual-CNN		✔				
	[161] 2019	1-D convolutional generative adversarial network		✔				
noise	[126] 2020	sliding-window based distance contrast					✔	

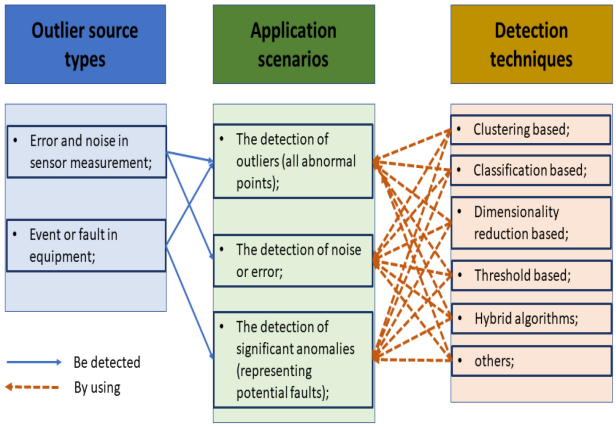


Fig. 11. Classification of outliers in IoT domain.

change in values and they can be either due to a drastic change in the environment (event or fault) or sensor malfunction (error). The work in [153] detects both error and event. The work in [140] reviews the new outlier detection approaches based on machine learning for IoT and the authors discuss the factors that contribute to outlier data, including environment factors, sensor factors, adversary factors, and event factors. Casado-Vara et al. [149] proposed a game theory approach to help eliminate (to a large extent) the possible thermal noise introduced by the sensors. The work in [150] directly treated outliers as anomalies.

2) *The Detection of Significant Anomalies (Representing Potential Faults):* When detecting the anomaly type of outliers, anomaly detection, and fault detection have gained a reputation in the IoT-SPM community. The frequent assumption is that data is noise-free and all outliers detected represent significant anomalies. The work in [155] exploits the PCA approach to diagnose possible faults early in heating, ventilation, and air conditioning (HVAC) systems to prevent further deterioration, which might lead to discomfort for the users, energy wastage, system unreliability, and equipment damage. The work in [157] utilizes autoregressive integrated moving average (ARIMA) to predict the sensor readings from a slitting machine and the points that have large prediction errors are identified as the failures to monitor the overall manufacturing process. More anomaly detection or fault detection methods have been developed for intelligent fault diagnosis in wind turbines [158], rotary machinery [159], [163], gears [164], motors [165], and so on. In recent years, with the rapid adoption of IoT sensor technologies, more diverse and advanced approaches [160], [161], [166], [167] have been proposed for various research areas and application domains.

3) *Discussion:* Most outlier detection methods produce both noise detection and anomaly detection results without making a clear distinction between them [141]. An increasing number of studies conduct research on anomaly detection, and some even simply refer to this as outlier detection [140]. In summary, all above methods fail to distinguish noise from anomalies. As pointed out in [155], a number of elements make anomaly detection very challenging and noise is one of them. Noise is inherent on real-world systems; therefore, it is important to consider this issue when deploying the IIoT system in real-word platforms. However, the work in [155] also mentioned that it may not be suitable to apply traditional noise reduction techniques to raw data

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TABLE V
COMPARATIVE TABLE OF MISSING VALUE IMPUTATION METHODS IN IOT

multiple variable imputation	
References	techniques
[169] 2008	replace with nearest neighbor
[170] 2005	association rule
[127] 2017	probabilistic matrix factorization
[171] 2021	generative adversarial imputation network (GAIN)
[178] 2019	LSTM-based deep learning imputation model
[179] 2021	variational Autoencoder (VAE), Neural Network with Random Weights (NNRW), Multiple Imputation by Chain Equations (MICE), Random Forest-based Imputation (missForest) and K-Nearest Neighbour (KNN)
[180] 2020	find the nearest neighbor using the defined similarity measures
[181] 2020	Bayesian maximum entropy (BME)
[182] 2015	missing value imputation based on context and linear mean (MCL), model of missing value imputation based on binary search (MBS), and model of missing value imputation based on Gaussian mixture model (MGI)
[183] 2020	Radial Basis Functions, Moving Least Squares (MLS), Adaptive Inverse Distance Weighted, KNN
[184] 2022	deep Hadamard autoencoders
[185] 2020	LSTM-based
univariate variables imputation	
[132] 2020	STLDecImp-based
[177] 2021	MPdist-based
[186] 2021	first-order Markov chain
[172] 2020	replaced by the mean value of the non-missing values

before the anomaly detection stage. To the best of the researcher's knowledge, not many researchers are aware of this problem. The work in [126] is the first attempt to explicitly introduce and describe noise detection in the presence of anomalies and provides an effective solution.

B. Missing Value Imputation

There are two main missing value imputation methods, namely, multiple variables-based and univariate variables-based methods. The first uses the entire set of variables to estimate the missing values, while the second imputes the missing values only using the nonmissing values in the same variable. Although [168] discusses the various strategies to handle the missing data, the survey on missing data imputation methods in IoT applications is completely missing. In this mini-review, we focus on the works that are specifically related to IoT applications. Table V presents a summary of missing value imputation methods in IoT.

1) *Multiple Variables-Based Imputation:* In the IoT paradigm, a massive number of sensors are installed and distributed around machines across the whole smart factory for health condition monitoring, and there are usually several sensors for each piece of equipment. Because of the unique characteristics of IoT monitoring systems, the sensor data from neighboring locations (geographically close to each other) should be highly correlated. Motivated by this observation, the work in [169] provides a spatial-temporal imputation method to replace the missing values with neighboring sensor readings. The work in [170] estimates the missing values using nonmissing values from the related sensors, which need to be determined after a series of analysis using an association rule mining algorithm, rather than simply using the values from neighboring sensors. In [127], probabilistic matrix factorization (PMF) is used to measure the similarity in data among neighboring sensors, then a k-means algorithm is applied to group them into different clusters. To estimate the missing values, the related sensors refer to the rest of the sensors from the same cluster. Generative adversarial imputation network (GAIN) was adopted in [171] to generate missing sensor measurements in IIoT.
2) *Univariate Variables-Based Imputation:* Common mode failures occur when a single event leads to the loss of data from multiple sensors. This is a commonly occurring scenario in IoT-enabled smart manufacturing monitoring systems. For example, when faced with network collapses and data transformation errors, a large number of sensors cannot deliver the measurement readings at the same time. In this case, the sensors with missing values might be not able to be imputed using the nonmissing values from other sensors since the related sensors lose the data as well during common mode failures. To deal with this problem, univariate variables-based imputation methods are required, which fill in the missing values by exploiting the historical observed nonmissing data in the same sensor. In industry, engineers prefer to use simple and straightforward methods, such as last observation carried forward (LOCF), simply imputing with a fixed value like mean value [172], regression [173], or interpolation imputation. There are also several advanced techniques reported in the literature, such as Kalman filter-based methods [174] and seasonal-trend decomposition procedures based on the loss (STL) decomposition-based imputation methods (STLDecImp) [175]. The work in [176] shows that for most imputation methods, the performance consistently decreases as the gap length becomes larger. Therefore, researchers have started to address the large gap issues. The work in [132] proposes an iterative imputation framework using multiple segmented gap iteration to provide more precise results for IIoT sensor data with large gaps. The work in [177] leveraged MPdist, which is more robust than the Euclidean distance (ED) or dynamic time warping (DTW), to complete the large missing gap(s) within univariate time series.

C. Sensor Selection

IIoT systems generally collect measurements from as many sensors as possible to have a better view of the sensor surroundings, so that sensor-equipped equipment can be monitored in real time. Intensive networked sensors are deployed to realize the seamless monitoring of smart manufacturing, generating a large number of features mixed with noisy and noninformative ones. Sensor selection is vitally important to prevent unwanted features from degrading the performance and decreasing efficiency. Before the prevalence of IIoT, feature selection has matured and developed into a certain stage. Sensor selection can be regarded as feature selection applied

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TABLE VI
COMPARATIVE TABLE OF FEATURE SELECTION METHODS FOR IOT-SPM APPLICATIONS

supervised methods	
References	techniques
[190], [191] 2014	MI-based methods
[192] 2017	MI-KPCA-based method
[128] 2018	MIC-based method
[193] 2014	GA-based method to choose the features that achieve maximum fault detection accuracy
[195] 2015	GA-based method to choose multiple feature sets for different types of faults
[17] 2022	ANOVA-based method
[204] 2018	Scikit-learn's Recursive feature elimination with cross-validation (RFECV) module to select features
[205] 2021	random forest algorithm to discover variable importance
unsupervised methods	
[190], [191], [196]–[200]	PCA-based
[201] 2019	CCA-based
[206] 2019	MI of each pair of features to remove the most redundant ones
[129] 2021	KNN-MST-based clustering method to divide the whole feature set into different types of faults
[207] 2020	correlation based hierarchical clustering and relative entropy through the measure of Kullback–Leibler divergence to generate feature subset with minimal redundancies

in IIoT. Related works on traditional feature selection methods can generally be classified into three categories: 1) the filter methods; 2) the wrapper methods; and 3) embedded methods [187]. The filter methods select features depending on their importance in terms of improving model performance. One of the commonly used criteria is the correlation between variables and the target, such as distance, consistency, dependency, information [188], [189], and so on. However, these heuristic filter methods are theoretically unjustified and the performance may not be satisfactory. Wrapper methods determine the optimal feature subset which yields the minimal classification or regression error. They achieve better results than filter methods, but with high computational complexity since they need to compare the predictive accuracy of all possible feature subsets. In summary, filter methods are fast (efficient) however the quality of the selected features is not very good (ineffective). In contrast, wrapper methods are effective but inefficient especially for large-scale data sets. In the following, we comprehensively review and qualitatively compare the feature selection algorithms developed in IIoT applications for IoT-SPM. A comparative summary can be found in Table VI.

1) *Supervised Learning:* The maximal information (MI)-based methods proposed in [190] and [191] select features by evaluating the relationship between each feature and the fault labels and choosing the most related ones. An MI-KPCA-based two-block monitoring scheme is provided in [192]. The work in [188] selects the optimal subset by solving an object programming problem where the object is to maximize the mutual information between candidate variables and the labels. To derive the optimal solution of such an NP-hard problem, a greedy feature filter is specifically proposed in that paper. The work in [128] presents a feature selection method based on the maximal information coefficient (MIC) for IoT applications. Ghosh et al. [193] proposed a stochastic optimization-based method using the genetic algorithm (GA) to identify an optimal feature subset that achieves maximum fault detection accuracy. Given a labeled data set, GA is utilized to search for a subset of sensor variables such that the monitoring model built on these variables has the best possible monitoring performance. This method can be classified as a wrapper-based feature selection approach and needs to fully explore the combinatorial solutions space with supervised learning to search for such a feature subset. The work in [194] utilized a GA-based approach to select an optimal sensor subset with maximum diagnosabilty for HVAC systems. The work in [195] modified the method proposed in [193] to accommodate multiple faults, and it is able to identify one optimal subset for each fault. The work in [17] leveraged analysis of variance (ANOVA) feature selection method for fault detection and diagnostics in a digital twin predictive maintenance framework.

Discussion: Previous approaches are well designed and work for traditional process monitoring. Most existing works are based on supervised learning. A common shortcoming of the supervised methods is the strict requirement of one-to-one labels which might be not easy to obtain in real industrial scenarios. When applied to IIoT, most existing solutions become infeasible due to the lack of labels and the new challenges brought about by BD. In GA, dealing with thousands of features is inefficient and time-consuming since it is not easy to traverse the thousands of features involved in combinatorial space to find the most optimal subset. In addition, for high-frequency and streaming industrial sensor data, it is very difficult and impractical to undertake manual labeling in industrial scenarios. The unavailability of labeled data prevents the use of all the existing supervised methods. This means there is a large research gap needing to be explored in terms of efficient unsupervised or weak-supervised learning.

2) *Unsupervised Learning:* For traditional process monitoring, most unsupervised learning methods are principal component (PC)-based. Several studies have been conducted on feature selection or division, such as multiblock or DPCA-based methods [190], [191], [196], [197], [198], [199], [200], and the distributed Canonical correlation analysis (CCA)-based method [201]. The work in [197] developed a DPCA model by constructing sub-blocks through different directions of PCs. Since these PCs are uncorrelated, the resulting sub-blocks can satisfy the diversity requirement of the block division method. The distributed monitoring scheme proposed in [202] employs PCA to derive four distinct and explicable subspaces. However, PCA does not explicitly reveal the cause-and-effect relationships [203]. PC-based methods partition the features based on variance and

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thus, they do not always select good features for fault detection, as features with a large variance can be independent of the intrinsic detection of the fault. Multiple sensors cannot be properly divided into sub-blocks suitable for different types of faults by only relying on variance. Thus, these types of methods are not suitable for feature selection in predicative maintenance-specific applications.

VII. Conclusion

In this article, we discussed the possible challenges and requirements of a general IoT architecture from a high level. For the device layer, more advanced sensors with various sensing capabilities need to be deployed in IIoT: 1) at all the monitoring points; 2) monitoring diverse physical metrics; and 3) providing seamless monitoring. With the large number of sensing technologies, the number of smart devices has grown greatly in recent years. This voluminous number of devices triggered the issues in terms of networking and communication. With regard to the application layer, this article gives a comprehensive analysis of data difficulties, including the low quality of raw data, the lack of industrial background, and new issues specifically introduced by the application of IoT-SPM. In light of the above discussions, we outline the key requirements which are deemed to be the most essential components for a precise and efficient IoT-SPM ecosystem. In what follows, current progress surrounding IoT-SPM and its components (also known as enabling technologies) are reviewed, based on which we identify a large gap of data quality in IoT. Finally, a reference IoT-SPM ecosystem is provided, which is integrated with a series of data processing operations motivated by data quality concerns. By listing the issues related to the data analysis layer, this article aims to raise people's awareness and draw attention to data difficulties in the IoT context. Regarding each issue, this article particularly provides a qualitative evaluation and comparison of IoT systems and data quality issues. The systems and approaches are compared and discussed carefully with respect to their advantages and drawbacks. Such reviews and comparisons also uncover open research issues that could be studied in the future. The discussion is in two sections, the first deals with future work on the data quality of IIoT and the second with IoT-SPM ecosystem migration.

Traditional artificial intelligence technologies cannot be used directly in IIoT fields and need huge transitions to accommodate IIoT applications. Most existing works discuss machine learning methods for fault detection applications, but there are very few comprehensive studies on IoT data quality that are specifically related or apply to IoT-SPM. To address IoT data quality issues, the requirements and directions in future research are as follows.

1) *Personalized Data Quality and Preprocessing Solutions:* Various challenges arise which are specific to the different data types and applications. In addition, in smart manufacturing, how to deploy IIoT to assist the industry process is subjective. Each customer has unique requirements and needs depending on the task purposes. Different users have varying versions of what "good" data should look like. We hope that more researchers focus on data quality issues. There is a need for an effective method or solution to be developed which is able to deal with personalized data quality management and software applications.
2) *Lightweight Solutions:* In IIoT environments, a real-time processing ability is one of the most basic requirements. So, it is desirable to design lightweight solutions that could be either embedded in the edge nodes near the physical objects or deployed in the cloud center.
3) *Metrics for Data Quality Assessment:* To the best of the author's knowledge, this research presents the first example where IoT application-specific data quality problems are defined and solved in a systematic way. We anticipate that this article will be used to show that data quality and data preprocessing for an IoT-based ecosystem in the manufacturing industry is an emerging data science area with great practical significance and research directions that need to be explored in the future. We expect that through this research, data quality issues will receive more attention and more sophisticated techniques for quality control and management will be developed in the coming years. The development of metrics for data quality is envisioned in the future to assess whether the data is good enough. Currently, there are no standard criteria to define the high-quality data since data quality is a subjective concept that varies depending on the intend use of the data [135], [208].

Moreover, it would be very beneficial if future works could elaborate a migration process to integrate an IoT-SPM ecosystem with the current supervisory control and data acquisition (SCADA) systems and address in detail the migration procedures and barriers/obstacles that may determine whether and to what extent the IoT-SPM ecosystem can be integrated into legacy industrial management, automation, and control architectures.

References

[1] F. Tao and Q. Qi, "Make more digital twins," *Nature*, vol. 573, no. 7775, pp. 490–491, 2019.
[2] J. D. Hochhalter et al., "Coupling damage-sensing particles to the digital twin concept," Nat. Aeronautics Space Admin., Langley Res. Center, Hampton, VA, USA, 2014.
[3] "Spatial services projects." Accessed: Dec. 29, 2021. [Online]. Available: https://www.spatial.nsw.gov.au/what_we_do/projects
[4] "Meet virtual Singapore, the city's 3D digital twin." Accessed: Dec. 29, 2021. [Online]. Available: https://govinsider.asia/digital-gov/meet-virtual-singapore-citys-3d-digital-twin/
[5] M. Biagioli and C. Galileo, *The Practice of Science in the Culture of Absolutism*, vol. 95. Chicago, IL, USA: Univ. Chicago Press, 1993.
[6] A. Fuller, Z. Fan, C. Day, and C. Barlow, "Digital twin: Enabling technologies, challenges and open research," *IEEE Access*, vol. 8, pp. 108952–108971, 2020.
[7] G. Mylonas, A. Kalogeras, G. Kalogeras, C. Anagnostopoulos, C. Alexakos, and L. Muñoz, "Digital twins from smart manufacturing to smart cities: A survey," *IEEE Access*, vol. 9, pp. 143222–143249, 2021.
[8] Y. Wu, K. Zhang, and Y. Zhang, "Digital twin networks: A survey," *IEEE Internet Things J.*, vol. 8, no. 18, pp. 13789–13804, Sep. 2021.
[9] F. Dembski, U. Wössner, M. Letzgus, M. Ruddat, and C. Yamu, "Urban digital twins for smart cities and citizens: The case study of herrenberg, germany," *Sustainability*, vol. 12, no. 6, p. 2307, 2020.

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[10] C. Verdouw, B. Tekinerdogan, A. Beulens, and S. Wolfert, "Digital twins in smart farming," *Agric. Syst.*, vol. 189, Apr. 2021, Art. no. 103046.

[11] P. O'Donovan, K. Leahy, K. Bruton, and D. T. O'Sullivan, "An industrial big data pipeline for data-driven analytics maintenance applications in large-scale smart manufacturing facilities," *J. Big Data*, vol. 2, no. 1, p. 25, 2015.

[12] I. Errandonea, S. Beltrán, and S. Arrizabalaga, "Digital twin for maintenance: A literature review," *Comput. Ind.*, vol. 123, Dec. 2020, Art. no. 103316.

[13] F. Tao, H. Zhang, A. Liu, and A. Y. Nee, "Digital twin in industry: State-of-the-art," *IEEE Trans. Ind. Informat.*, vol. 15, no. 4, pp. 2405–2415, Apr. 2019.

[14] W. Kritzinger, M. Karner, G. Traar, J. Henjes, and W. Sihn, "Digital twin in manufacturing: A categorical literature review and classification," *IFAC-PapersOnLine*, vol. 51, no. 11, pp. 1016–1022, 2018.

[15] G. Falekas and A. Karlis, "Digital twin in electrical machine control and predictive maintenance: State-of-the-art and future prospects," *Energies*, vol. 14, no. 18, p. 5933, 2021.

[16] M. Liu, S. Fang, H. Dong, and C. Xu, "Review of digital twin about concepts, technologies, and industrial applications," *J. Manuf. Syst.*, vol. 58, pp. 346–361, Jan. 2021.

[17] H. H. Hosamo, P. R. Svennevig, K. Svidt, D. Han, and H. K. Nielsen, "A digital twin predictive maintenance framework of air handling units based on automatic fault detection and diagnostics," *Energy Build.*, vol. 261, Apr. 2022, Art. no. 111988.

[18] S. Venkatesan, K. Manickavasagam, N. Tengenkai, and N. Vijayalakshmi, "Health monitoring and prognosis of electric vehicle motor using intelligent-digital twin," *IET Elect. Power Appl.*, vol. 13, no. 9, pp. 1328–1335, 2019.

[19] J. Wang, L. Ye, R. X. Gao, C. Li, and L. Zhang, "Digital twin for rotating machinery fault diagnosis in smart manufacturing," *Int. J. Prod. Res.*, vol. 57, no. 12, pp. 3920–3934, 2019.

[20] R. van Dinter, B. Tekinerdogan, and C. Catal, "Predictive maintenance using digital twins: A systematic literature review," *Inf. Softw. Technol.*, vol. 151, Nov. 2022, Art. no. 107008.

[21] I. Yaqoob et al., "Internet of things architecture: Recent advances, taxonomy, requirements, and open challenges," *IEEE Wireless Commun.*, vol. 24, no. 3, pp. 10–16, Jun. 2017.

[22] P. P. Ray, "A survey on Internet of Things architectures," *J. King Saud Univ. Comput. Inf. Sci.*, vol. 30, no. 3, pp. 291–319, 2018.

[23] N. C. Luong, D. T. Hoang, P. Wang, D. Niyato, D. I. Kim, and Z. Han, "Data collection and wireless communication in Internet of Things (IoT) using economic analysis and pricing models: A survey," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 4, pp. 2546–2590, 4th Quart., 2016.

[24] K. Xu, Y. Qu, and K. Yang, "A tutorial on the Internet of Things: From a heterogeneous network integration perspective," *IEEE Netw.*, vol. 30, no. 2, pp. 102–108, Mar./Apr. 2016.

[25] L. Jiang, L. Da Xu, H. Cai, Z. Jiang, F. Bu, and B. Xu, "An IoT-oriented data storage framework in cloud computing platform," *IEEE Trans. Ind. Informat.*, vol. 10, no. 2, pp. 1443–1451, May 2014.

[26] H. Cai, B. Xu, L. Jiang, and A. V. Vasilakos, "IoT-based big data storage systems in cloud computing: Perspectives and challenges," *IEEE Internet Things J.*, vol. 4, no. 1, pp. 75–87, Feb. 2017.

[27] J. Singh, T. Pasquier, J. Bacon, H. Ko, and D. Eyers, "Twenty security considerations for cloud-supported Internet of Things," *IEEE Internet Things J.*, vol. 3, no. 3, pp. 269–284, Jun. 2016.

[28] Y. Yang, L. Wu, G. Yin, L. Li, and H. Zhao, "A survey on security and privacy issues in Internet-of-Things," *IEEE Internet Things J.*, vol. 4, no. 5, pp. 1250–1258, Oct. 2017.

[29] Y. Sun, H. Song, A. J. Jara, and R. Bie, "Internet of Things and big data analytics for smart and connected communities," *IEEE Access*, vol. 4, pp. 766–773, 2016.

[30] L. Wang and R. Ranjan, "Processing distributed Internet of Things data in clouds," *IEEE Cloud Comput.*, vol. 2, no. 1, pp. 76–80, Jan./Feb. 2015.

[31] M. S. Hossain and G. Muhammad, "Cloud-assisted Industrial Internet of Things (IIoT)– Enabled framework for health monitoring," *Comput. Netw.*, vol. 101, pp. 192–202, Jun. 2016.

[32] L. Atzori, A. Iera, and G. Morabito, "The Internet of Things: A survey," *Comput. Netw.*, vol. 54, no. 15, pp. 2787–2805, 2010.

[33] L. Da Xu, W. He, and S. Li, "Internet of Things in industries: A survey," *IEEE Trans. Ind. Informat.*, vol. 10, no. 4, pp. 2233–2243, Nov. 2014.

[34] R. Van Kranenburg, *The Internet of Things: A Critique of Ambient Technology and the All-Seeing Network of RFID*, Inst. Netw. Cult., Amsterdam, The Netherlands, 2008.

[35] A. Zanella, N. Bui, A. Castellani, L. Vangelista, and M. Zorzi, "Internet of Things for smart cities," *IEEE Internet Things J.*, vol. 1, no. 1, pp. 22–32, Feb. 2014.

[36] C. Stergiou, K. E. Psannis, B.-G. Kim, and B. Gupta, "Secure integration of IoT and cloud computing," *Future Gener. Comput. Syst.*, vol. 78, pp. 964–975, Jan. 2018.

[37] J. Lin, W. Yu, N. Zhang, X. Yang, H. Zhang, and W. Zhao, "A survey on Internet of Things: Architecture, enabling technologies, security and privacy, and applications," *IEEE Internet Things J.*, vol. 4, no. 5, pp. 1125–1142, Oct. 2017.

[38] T. S. Nikoui, A. M. Rahmani, A. Balador, and H. H. S. Javadi, "Internet of Things architecture challenges: A systematic review," *Int. J. Commun. Syst.*, vol. 34, no. 4, p. e4678, 2021.

[39] F. Tao, J. Cheng, and Q. Qi, "IiHub: An Industrial Internet-of-Things hub toward smart manufacturing based on cyber-physical system," *IEEE Trans. Ind. Informat.*, vol. 14, no. 5, pp. 2271–2280, May 2018.

[40] M. Wollschlaeger, T. Sauter, and J. Jasperneite, "The future of industrial communication: Automation networks in the era of the Internet of Things and industry 4.0," *IEEE Ind. Electron. Mag.*, vol. 11, no. 1, pp. 17–27, Mar. 2017.

[41] L. Monostori et al., "Cyber-physical systems in manufacturing," *Cirp Ann.*, vol. 65, no. 2, pp. 621–641, 2016.

[42] A. Burg, A. Chattopadhyay, and K.-Y. Lam, "Wireless communication and security issues for cyber–physical systems and the Internet-of-Things," *Proc. IEEE*, vol. 106, no. 1, pp. 38–60, Jan. 2018.

[43] Z. Gao, C. Cecati, and S. X. Ding, "A survey of fault diagnosis and fault-tolerant techniques–Part I: Fault diagnosis with model-based and signal-based approaches," *IEEE Trans. Ind. Electron.*, vol. 62, no. 6, pp. 3757–3767, Jun. 2015.

[44] A. M. S. Osman, "A novel big data analytics framework for smart cities," *Future Gener. Comput. Syst.*, vol. 91, pp. 620–633, Feb. 2019.

[45] C.-W. Tsai, C.-F. Lai, H.-C. Chao, and A. V. Vasilakos, "Big data analytics: A survey," *J. Big Data*, vol. 2, no. 1, p. 21, 2015.

[46] "Hadoop apache." Accessed: Feb. 21, 2021. [Online]. Available: https://hadoop.apache.org/

[47] P. Hu, S. Dhelim, H. Ning, and T. Qiu, "Survey on fog computing: Architecture, key technologies, applications and open issues," *J. Netw. Comput. Appl.*, vol. 98, pp. 27–42, Nov. 2017.

[48] W. Yu, T. Dillon, F. Mostafa, W. Rahayu, and Y. Liu, "A global manufacturing big data ecosystem for fault detection in predictive maintenance," *IEEE Trans. Ind. Informat.*, vol. 16, no. 1, pp. 183–192, Jan. 2020.

[49] X. Sun and N. Ansari, "EdgeIoT: Mobile edge computing for the Internet of Things," *IEEE Commun. Mag.*, vol. 54, no. 12, pp. 22–29, Dec. 2016.

[50] A. Al-Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, and M. Ayyash, "Internet of Things: A survey on enabling technologies, protocols, and applications," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 4, pp. 2347–2376, 4th Quart., 2015.

[51] O. Said and M. Masud, "Towards Internet of Things: Survey and future vision," *Int. J. Comput. Netw.*, vol. 5, no. 1, pp. 1–17, 2013.

[52] L. Lao, Z. Li, S. Hou, B. Xiao, S. Guo, and Y. Yang, "A survey of IoT applications in blockchain systems: Architecture, consensus, and traffic modeling," *ACM Comput. Surveys*, vol. 53, no. 1, pp. 1–32, 2020.

[53] T. Qiu, J. Chi, X. Zhou, Z. Ning, M. Atiquzzaman, and D. O. Wu, "Edge computing in Industrial Internet of Things: Architecture, advances and challenges," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 4, pp. 2462–2488, 4th Quart., 2020.

[54] J. Maktoubian and K. Ansari, "An IoT architecture for preventive maintenance of medical devices in healthcare organizations," *Health Technol.*, vol. 9, no. 3, pp. 233–243, 2019.

[55] A. S. Pillai, G. S. Chandraprasad, A. S. Khwaja, and A. Anpalagan, "A service oriented IoT architecture for disaster preparedness and forecasting system," *Internet Things*, vol. 14, Jun. 2021, Art. no. 100076.

[56] C. Liu, F. Chen, C. Zhao, T. Wang, C. Zhang, and Z. Zhang, "IPv6-based architecture of community medical Internet of Things," *IEEE Access*, vol. 6, pp. 7897–7910, 2018.

[57] S. Krčo, B. Pokrić, and F. Carrez, "Designing IoT architecture(s): A european perspective," in *Proc. IEEE World Forum Internet Things (WF-IoT)*, 2014, pp. 79–84.

[58] R. T. Tiburski, L. A. Amaral, E. De Matos, and F. Hessel, "The importance of a standard securit y archit ecture for SOA-based IoT middleware," *IEEE Commun. Mag.*, vol. 53, no. 12, pp. 20–26, Dec. 2015.

Authorized licensed use limited to: Pontificia Universidad Javeriana. Downloaded on October 14,2026 at 23:22:55 UTC from IEEE Xplore. Restrictions apply.

[59] D. Minoli, K. Sohraby, and B. Occhiogrosso, "IoT considerations, requirements, and architectures for smart buildings—Energy optimization and next-generation building management systems," *IEEE Internet Things J.*, vol. 4, no. 1, pp. 269–283, Feb. 2017.

[60] M. Weyrich and C. Ebert, "Reference architectures for the Internet of Things," *IEEE Softw.*, vol. 33, no. 1, pp. 112–116, Jan./Feb. 2016.

[61] Y. He, J. Guo, and X. Zheng, "From surveillance to digital twin: Challenges and recent advances of signal processing for Industrial Internet of Things," *IEEE Signal Process. Mag.*, vol. 35, no. 5, pp. 120–129, Sep. 2018.

[62] Z. Xu, L. Chen, C. Chen, and X. Guan, "Joint clustering and routing design for reliable and efficient data collection in large-scale wireless sensor networks," *IEEE Internet Things J.*, vol. 3, no. 4, pp. 520–532, Aug. 2016.

[63] M. Dong, K. Ota, and A. Liu, "RMER: Reliable and energy-efficient data collection for large-scale wireless sensor networks," *IEEE Internet Things J.*, vol. 3, no. 4, pp. 511–519, Aug. 2016.

[64] T. Wang, L. Qiu, A. K. Sangaiah, G. Xu, and A. Liu, "Energy-efficient and trustworthy data collection protocol based on mobile fog computing in Internet of Things," *IEEE Trans. Ind. Informat.*, vol. 16, no. 5, pp. 3531–3539, May 2020.

[65] Y. Duan, W. Li, X. Fu, Y. Luo, and L. Yang, "A methodology for reliability of WSN based on software defined network in adaptive industrial environment," *IEEE/CAA J. Automatica Sinica*, vol. 5, no. 1, pp. 74–82, Jan. 2018.

[66] J. Wan, J. Yang, S. Wang, D. Li, P. Li, and M. Xia, "Cross-network fusion and scheduling for heterogeneous networks in smart factory," *IEEE Trans. Ind. Informat.*, vol. 16, no. 9, pp. 6059–6068, Sep. 2020.

[67] N. Chervyakov, M. Babenko, A. Tchernykh, N. Kucherov, V. Miranda-López, and J. M. Cortés-Mendoza, "AR-RRNS: Configurable reliable distributed data storage systems for Internet of Things to ensure security," *Future Gener. Comput. Syst.*, vol. 92, pp. 1080–1092, Mar. 2019.

[68] T. Wang, L. Qiu, A. K. Sangaiah, A. Liu, M. Z. A. Bhuiyan, and Y. Ma, "Edge-computing-based trustworthy data collection model in the Internet of Things," *IEEE Internet Things J.*, vol. 7, no. 5, pp. 4218–4227, May 2020.

[69] E. Luo, M. Z. A. Bhuiyan, G. Wang, M. A. Rahman, J. Wu, and M. Atiquzzaman, "Privacyprotector: Privacy-protected patient data collection in IoT-based healthcare systems," *IEEE Commun. Mag.*, vol. 56, no. 2, pp. 163–168, Feb. 2018.

[70] H. Tao, M. Z. A. Bhuiyan, A. N. Abdalla, M. M. Hassan, J. M. Zain, and T. Hayajneh, "Secured data collection with hardware-based ciphers for IoT-based healthcare," *IEEE Internet Things J.*, vol. 6, no. 1, pp. 410–420, Feb. 2019.

[71] T. Li, W. Liu, T. Wang, Z. Ming, X. Li, and M. Ma, "Trust data collections via vehicles joint with unmanned aerial vehicles in the smart Internet of Things," *Trans. Emerg. Telecommun. Technol.*, vol. 33, no. 5, p. e3956, 2022.

[72] H. Landaluce, L. Arjona, A. Perallos, F. Falcone, I. Angulo, and F. Muralter, "A review of IoT sensing applications and challenges using RFID and wireless sensor networks," *Sensors*, vol. 20, no. 9, p. 2495, 2020.

[73] Q. Meng and S. Zhu, "Developing IoT sensing system for construction-induced vibration monitoring and impact assessment," *Sensors*, vol. 20, no. 21, p. 6120, 2020.

[74] Y. Cui, F. Liu, X. Jing, and J. Mu, "Integrating sensing and communications for ubiquitous IoT: Applications, trends, and challenges," *IEEE Netw.*, vol. 35, no. 5, pp. 158–167, Sep./Oct. 2021.

[75] K. Lin, Y. Li, Q. Zhang, and G. Fortino, "Ai-driven collaborative resource allocation for task execution in 6G-enabled massive IoT," *IEEE Internet Things J.*, vol. 8, no. 7, pp. 5264–5273, Apr. 2021.

[76] X. Li and L. Da Xu, "A review of Internet of Things—Resource allocation," *IEEE Internet Things J.*, vol. 8, no. 11, pp. 8657–8666, Jun. 2021.

[77] H. Zhang, Y. Xiao, S. Bu, D. Niyato, F. R. Yu, and Z. Han, "Computing resource allocation in three-tier IoT fog networks: A joint optimization approach combining stackelberg game and matching," *IEEE Internet Things J.*, vol. 4, no. 5, pp. 1204–1215, Oct. 2017.

[78] X. Xiong, K. Zheng, L. Lei, and L. Hou, "Resource allocation based on deep reinforcement learning in IoT edge computing," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 6, pp. 1133–1146, Jun. 2020.

[79] H. Li, K. Ota, and M. Dong, "Learning iot in edge: Deep learning for the Internet of Things with edge computing," *IEEE Netw.*, vol. 32, no. 1, pp. 96–101, Jan./Feb. 2018.

[80] L. Li, K. Ota, and M. Dong, "Deep learning for smart industry: Efficient manufacture inspection system with fog computing," *IEEE Trans. Ind. Informat.*, vol. 14, no. 10, pp. 4665–4673, Oct. 2018.

[81] T. P. Carvalho, F. A. Soares, R. Vita, R. D. P. Francisco, J. P. Basto, and S. G. Alcalá, "A systematic literature review of machine learning methods applied to predictive maintenance," *Comput. Ind. Eng.*, vol. 137, Nov. 2019, Art. no. 106024.

[82] W. Zhang, D. Yang, and H. Wang, "Data-driven methods for predictive maintenance of industrial equipment: A survey," *IEEE Syst. J.*, vol. 13, no. 3, pp. 2213–2227, Sep. 2019.

[83] M. S. Mahdavinejad, M. Rezvan, M. Barekatain, P. Adibi, P. Barnaghi, and A. P. Sheth, "Machine learning for Internet of Things data analysis: A survey," *Digit. Commun. Netw.*, vol. 4, no. 3, pp. 161–175, 2018.

[84] S. Namuduri, B. N. Narayanan, V. S. P. Davuluru, L. Burton, and S. Bhansali, "Deep learning methods for sensor based predictive maintenance and future perspectives for electrochemical sensors," *J. Electrochem. Soc.*, vol. 167, no. 3, 2020, Art. no. 37552.

[85] O. Serradilla, E. Zugasti, J. Rodriguez, and U. Zurutuza, "Deep learning models for predictive maintenance: A survey, comparison, challenges and prospects," *Appl. Intell.*, vol. 50, pp. 10934–10964, Aug. 2022.

[86] F. Tao, Y. Cheng, L. Da Xu, L. Zhang, and B. H. Li, "CCIoT-CMfg: Cloud computing and Internet of Things-based cloud manufacturing service system," *IEEE Trans. Ind. Informat.*, vol. 10, no. 2, pp. 1435–1442, May 2014.

[87] J. Wan et al., "A manufacturing big data solution for active preventive maintenance," *IEEE Trans. Ind. Informat.*, vol. 13, no. 4, pp. 2039–2047, Aug. 2017.

[88] M. M. Rathore, A. Ahmad, A. Paul, and S. Rho, "Urban planning and building smart cities based on the Internet of Things using big data analytics," *Comput. Netw.*, vol. 101, pp. 63–80, Jun. 2016.

[89] O. Elijah, T. A. Rahman, I. Orikumhi, C. Y. Leow, and M. N. Hindia, "An overview of Internet of Things (IoT) and data analytics in agriculture: Benefits and challenges," *IEEE Internet Things J.*, vol. 5, no. 5, pp. 3758–3773, Oct. 2018.

[90] M. S. Farooq, S. Riaz, A. Abid, T. Umer, and Y. B. Zikria, "Role of IoT technology in agriculture: A systematic literature review," *Electronics*, vol. 9, no. 2, p. 319, 2020.

[91] E. Spanò, L. Niccolini, S. Di Pascoli, and G. Iannacconeluca, "Last-meter smart grid embedded in an Internet-of-Things platform," *IEEE Trans. Smart Grid*, vol. 6, no. 1, pp. 468–476, Jan. 2015.

[92] F. Zantalis, G. Koulouras, S. Karabetsos, and D. Kandris, "A review of machine learning and IoT in smart transportation," *Future Internet*, vol. 11, no. 4, p. 94, 2019.

[93] E. Navarro, N. Costa, and A. Pereira, "A systematic review of IoT solutions for smart farming," *Sensors*, vol. 20, no. 15, p. 4231, 2020.

[94] A. D. Boursianis et al., "Internet of Things (IoT) and agricultural unmanned aerial vehicles (UAVs) in smart farming: A comprehensive review," *Internet Things*, vol. 18, May 2020, Art. no. 100187.

[95] H. Yang, S. Kumara, S. T. Bukkapatnam, and F. Tsung, "The Internet of Things for smart manufacturing: A review," *IISE Trans.*, vol. 51, no. 11, pp. 1190–1216, 2019.

[96] Y. Qu, X. Ming, Z. Liu, X. Zhang, and Z. Hou, "Smart manufacturing systems: State of the art and future trends," *Int. J. Adv. Manuf. Technol.*, vol. 103, no. 9, pp. 3751–3768, 2019.

[97] Q. Wang, X. Zhu, Y. Ni, L. Gu, and H. Zhu, "Blockchain for the IoT and industrial IoT: A review," *Internet Things*, vol. 10, Jun. 2020 Art. no. 100081.

[98] I. Mistry, S. Tanwar, S. Tyagi, and N. Kumar, "Blockchain for 5G-enabled IoT for industrial automation: A systematic review, solutions, and challenges," *Mech. Syst. Signal Process.*, vol. 135, Jan. 2020, Art. no. 106382.

[99] M. Fahim and A. Sillitti, "Anomaly detection, analysis and prediction techniques in IoT environment: A systematic literature review," *IEEE Access*, vol. 7, pp. 81664–81681, 2019.

[100] S. A. Dehkordi, K. Farajzadeh, J. Rezazadeh, R. Farahbakhsh, K. Sandrasegaran, and M. A. Dehkordi, "A survey on data aggregation techniques in IoT sensor networks," *Wireless Netw.*, vol. 26, no. 2, pp. 1243–1263, 2020.

[101] V. Hassija, V. Chamola, V. Saxena, D. Jain, P. Goyal, and B. Sikdar, "A survey on IoT security: Application areas, security threats, and solution architectures," *IEEE Access*, vol. 7, pp. 82721–82743, 2019.

[102] I. Sittón-Candanedo, R. S. Alonso, J. M. Corchado, S. Rodríguez-González, and R. Casado-Vara, "A review of edge computing reference architectures and a new global edge proposal," *Future Gener. Comput. Syst.*, vol. 99, pp. 278–294, Oct. 2019.

[103] S. Ayad, L. S. Terrissa, and N. Zerhouni, "An IoT approach for a smart maintenance," in *Proc. Int. Conf. Adv. Syst. Elect. Technol. (IC_ASET)*, 2018, pp. 210–214.

Authorized licensed use limited to: Pontificia Universidad Javeriana. Downloaded on October 14,2026 at 23:22:55 UTC from IEEE Xplore. Restrictions apply.

[104] H. Lee, "Framework and development of fault detection classification using IoT device and cloud environment," *J. Manuf. Syst.*, vol. 43, pp. 257–270, Apr. 2017.

[105] E. Ranjbar, R. G. Sedehi, M. Rashidi, and A. A. Suratgar, "Design of an IoT-based system for smart maintenance of medical equipment," in *Proc. 3rd Int. Conf. Internet Things Appl. (IoT)*, 2019, pp. 1–12.

[106] I. Niyonambaza, M. Zennaro, and A. Uwitonze, "Predictive maintenance (PdM) structure using Internet of Things (IoT) for mechanical equipment used into hospitals in rwanda," *Future Internet*, vol. 12, no. 12, p. 224, 2020.

[107] F. Civerchia, S. Bocchino, C. Salvadori, E. Rossi, L. Maggiani, and M. Petracca, "Industrial Internet of Things monitoring solution for advanced predictive maintenance applications," *J. Ind. Inf. Integr.*, vol. 7, pp. 4–12, Sep. 2017.

[108] R. Y. Zhong, L. Wang, and X. Xu, "An IoT-enabled real-time machine status monitoring approach for cloud manufacturing," *Procedia CIRP*, vol. 63, pp. 709–714, May 2017.

[109] W. Li and S. Kara, "Methodology for monitoring manufacturing environment by using wireless sensor networks (WSN) and the Internet of Things (IoT)," *Procedia CIRP*, vol. 61, pp. 323–328, Jan. 2017.

[110] M. Canizo, A. Conde, S. Charramendieta, R. Minon, R. G. Cid-Fuentes, and E. Onieva, "Implementation of a large-scale platform for cyber-physical system real-time monitoring," *IEEE Access*, vol. 7, pp. 52455–52466, 2019.

[111] B. Chen, J. Wan, A. Celesti, D. Li, H. Abbas, and Q. Zhang, "Edge computing in IoT-based manufacturing," *IEEE Commun. Mag.*, vol. 56, no. 9, pp. 103–109, Sep. 2018.

[112] D. Wu et al., "A fog computing-based framework for process monitoring and prognosis in cyber-manufacturing," *J. Manuf. Syst.*, vol. 43, pp. 25–34, Apr. 2017.

[113] W. Yu, Y. Liu, T. S. Dillon, and W. Rahayu, "Edge computing-assisted IoT framework with an autoencoder for fault detection in manufacturing predictive maintenance," *IEEE Trans. Ind. Informat.*, early access, May 30, 2022, doi: 10.1109/TII.2022.3178732.

[114] S. Ayvaz and K. Alpay, "Predictive maintenance system for production lines in manufacturing: A machine learning approach using IoT data in real-time," *Expert Syst. Appl.*, vol. 173, Jul. 2021, Art. no. 114598.

[115] P. Moens et al., "Scalable fleet monitoring and visualization for smart machine maintenance and industrial IoT applications," *Sensors*, vol. 20, no. 15, p. 4308, 2020.

[116] H. Ghallab, H. Fahmy, and M. Nasr, "Detection outliers on Internet of Things using big data technology," *Egyptian Informat. J.*, vol. 21, no. 3, pp. 131–138, 2020.

[117] M. Janssen, S. Luthra, S. Mangla, N. P. Rana, and Y. K. Dwivedi, "Challenges for adopting and implementing IoT in smart cities: An integrated MICMAC-ISM approach," *Internet Res.*, vol. 29 no. 6, pp. 1589–1616, 2019.

[118] J. Byabazaire, G. O'Hare, and D. Delaney, "Data quality and trust: A perception from shared data in IoT," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, 2020, pp. 1–6.

[119] N. Côrte-Real, P. Ruivo, and T. Oliveira, "Leveraging Internet of Things and big data analytics initiatives in european and american firms: Is data quality a way to extract business value?" *Inf. Manag.*, vol. 57, no. 1, 2020, Art. no. 103141.

[120] X. Liu et al., "Enhancing veracity of iot generated big data in decision making," in *Proc. IEEE Int. Conf. Pervasive Comput. Commun. Workshops (PerCom Workshops)*, 2018, pp. 149–154.

[121] A. Karkouch, H. Mousannif, H. Al Moatassime, and T. Noel, "Data quality in Internet of Things: A state-of-the-art survey," *J. Netw. Comput. Appl.*, vol. 73, pp. 57–81, Sep. 2016.

[122] M. Chen, S. Mao, and Y. Liu, "Big data: A survey," *Mobile Netw. Appl.*, vol. 19, no. 2, pp. 171–209, 2014.

[123] Y. Cui, S. Kara, and K. C. Chan, "Manufacturing big data ecosystem: A systematic literature review," *Robot. Comput. Integr. Manuf.*, vol. 62, Apr. 2020, Art. no. 101861.

[124] A. Nawaz et al., "Edge computing to secure IoT data ownership and trade with the ethereum blockchain," *Sensors*, vol. 20, no. 14, p. 3965, 2020.

[125] V. Janeček, "Ownership of personal data in the Internet of Things," *Comput. Law Secur. Rev.*, vol. 34, no. 5, pp. 1039–1052, 2018.

[126] Y. Liu, T. Dillon, W. Yu, W. Rahayu, and F. Mostafa, "Noise removal in the presence of significant anomalies for Industrial IoT sensor data in manufacturing," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 7084–7096, Aug. 2020.

[127] B. Fekade, T. Maksymyuk, M. Kyryk, and M. Jo, "Probabilistic recovery of incomplete sensed data in IoT," *IEEE Internet Things J.*, vol. 5, no. 4, pp. 2282–2292, Aug. 2018.

[128] G. Sun, J. Li, J. Dai, Z. Song, and F. Lang, "Feature selection for IoT based on maximal information coefficient," *Future Gener. Comput. Syst.*, vol. 89, pp. 606–616, Dec. 2018.

[129] Y. Liu, W. Yu, T. Dillon, W. Rahayu, and M. Li, "Empowering IoT predictive maintenance solutions with AI: A distributed system for manufacturing plant-wide monitoring," *IEEE Trans. Ind. Informat.*, vol. 18, no. 2, pp. 1345–1354, Feb. 2022.

[130] W. Yu, T. Dillon, F. Mostafa, W. Rahayu, and Y. Liu, "Implementation of industrial cyber physical system: Challenges and solutions," in *Proc. IEEE Int. Conf. Ind. Cyber Phys. Syst. (ICPS)*, 2019, pp. 173–178.

[131] W. Yu, Y. Liu, T. Dillon, W. Rahayu, and F. Mostafa, "An integrated framework for health state monitoring in a smart factory employing IoT and big data techniques," *IEEE Internet Things J.*, vol. 9, no. 3, pp. 2443–2454, Feb. 2022.

[132] Y. Liu, T. Dillon, W. Yu, W. Rahayu, and F. Mostafa, "Missing value imputation for Industrial IoT sensor data with large gaps," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 6855–6867, Aug. 2020.

[133] M. M. Farooqi, H. A. Khattak, and M. Imran, "Data quality techniques in the Internet of Things: Random forest regression," in *Proc. 14th Int. Conf. Emerg. Technol. (ICET)*, 2018, pp. 1–4.

[134] C. Liu, P. Nitschke, S. P. Williams, and D. Zowghi, "Data quality and the Internet of Things," *Computing*, vol. 102, no. 2, pp. 573–599, 2020.

[135] A. A. Alwan, M. A. Ciupala, A. J. Brimicombe, S. A. Ghorashi, A. Baravalle, and P. Falcarin, "Data quality challenges in large-scale cyber-physical systems: A systematic review," *Inf. Syst.*, vol. 105, Mar. 2022, Art. no. 101951.

[136] T. Banerjee and A. Sheth, "IoT quality control for data and application needs," *IEEE Intell. Syst.*, vol. 32, no. 2, pp. 68–73, Mar./Apr. 2017.

[137] T.-B. Ottosen and P. Kumar, "Outlier detection and gap filling methodologies for low-cost air quality measurements," *Environ. Sci. Process. Impacts*, vol. 21, no. 4, pp. 701–713, 2019.

[138] A. Cook, G. Mısırlı, and Z. Fan, "Anomaly detection for IoT time-series data: A survey," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 6481–6494, Jul. 2020.

[139] A. Ayadi, O. Ghorbel, A. M. Obeid, and M. Abid, "Outlier detection approaches for wireless sensor networks: A survey," *Comput. Netw.*, vol. 129, pp. 319–333, Dec. 2017.

[140] J. Jiang, G. Han, L. Liu, L. Shu, M. Guizani, "Outlier detection approaches based on machine learning in the Internet-of-Things," *IEEE Wireless Commun.*, vol. 27, no. 3, pp. 53–59, Jun. 2020.

[141] S. Basu and M. Meckesheimer, "Automatic outlier detection for time series: An application to sensor data," *Knowl. Inf. Syst.*, vol. 11, no. 2, pp. 137–154, 2007.

[142] M. Kang, M. R. Islam, J. Kim, J.-M. Kim, and M. Pecht, "A hybrid feature selection scheme for reducing diagnostic performance deterioration caused by outliers in data-driven diagnostics," *IEEE Trans. Ind. Electron.*, vol. 63, no. 5, pp. 3299–3310, May 2016.

[143] M. Gupta, J. Gao, C. C. Aggarwal, and J. Han, "Outlier detection for temporal data: A survey," *IEEE Trans. Knowl. Data Eng.*, vol. 26, no. 9, pp. 2250–2267, Sep. 2014.

[144] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Comput. Surveys*, vol. 41, no. 3, pp. 1–58, 2009.

[145] D. J. Hill and B. S. Minsker, "Anomaly detection in streaming environmental sensor data: A data-driven modeling approach," *Environ. Model. Softw.*, vol. 25, no. 9, pp. 1014–1022, 2010.

[146] E. Bigdeli, M. Mohammadi, B. Raahemi, and S. Matwin, "A fast and noise resilient cluster-based anomaly detection," *Pattern Anal. Appl.*, vol. 20, no. 1, pp. 183–199, 2017.

[147] A. M. Souza and J. R. Amazonas, "An outlier detect algorithm using big data processing and Internet of Things architecture," *Procedia Comput. Sci.*, vol. 52, pp. 1010–1015, 2015.

[148] M. Zhang, X. Li, and L. Wang, "An adaptive outlier detection and processing approach towards time series sensor data," *IEEE Access*, vol. 7, pp. 175192–175212, 2019.

[149] R. Casado-Vara, F. Prieto-Castrillo, and J. M. Corchado, "A game theory approach for cooperative control to improve data quality and false data detection in WSN," *Int. J. Robust Nonlinear Control*, vol. 28, no. 16, pp. 5087–5102, 2018.

[150] X. Deng, P. Jiang, X. Peng, and C. Mi, "An intelligent outlier detection method with one class support tucker machine and genetic algorithm toward big sensor data in Internet of Things," *IEEE Trans. Ind. Electron.*, vol. 66, no. 6, pp. 4672–4683, Jun. 2019.

[151] T. Yu, X. Wang, and A. Shami, "Recursive principal component analysis-based data outlier detection and sensor data aggregation in IoT systems," *IEEE Internet Things J.*, vol. 4, no. 6, pp. 2207–2216, Dec. 2017.

Authorized licensed use limited to: Pontificia Universidad Javeriana. Downloaded on October 14,2026 at 23:22:55 UTC from IEEE Xplore. Restrictions apply.

[152] H. S. Emadi and S. M. Mazinani, "A novel anomaly detection algorithm using DBSCAN and SVM in wireless sensor networks," *Wireless Pers. Commun.*, vol. 98, no. 12, pp. 2025–2035, 2018.

[153] N. Nesa, T. Ghosh, and I. Banerjee, "Outlier detection in sensed data using statistical learning models for IoT," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, 2018, pp. 1–6.

[154] R. Zhu et al., "KNN-based approximate outlier detection algorithm over IoT streaming data," *IEEE Access*, vol. 8, pp. 42749–42759, 2020.

[155] A. Beghi, R. Brignoli, L. Cecchinato, G. Menegazzo, M. Rampazzo, and F. Simmini, "Data-driven fault detection and diagnosis for HVAC water chillers," *Control Eng. Pract.*, vol. 53, pp. 79–91, Aug. 2016.

[156] Y. Liu, Z. Pang, M. Karlsson, and S. Gong, "Anomaly detection based on machine learning in IoT-based vertical plant wall for indoor climate control," *Build. Environ.*, vol. 183, Oct. 2020, Art. no. 107212.

[157] A. Kanawaday and A. Sane, "Machine learning for predictive maintenance of industrial machines using IoT sensor data," in *Proc. 8th IEEE Int. Conf. Softw. Eng. Service Sci. (ICSESS)*, 2017, pp. 87–90.

[158] K. Leahy, R. L. Hu, I. C. Konstantakopoulos, C. J. Spanos, and A. M. Agogino, "Diagnosing wind turbine faults using machine learning techniques applied to operational data," in *Proc. IEEE Int. Conf. Prognost. Health Manag. (ICPHM)*, 2016, pp. 1–8.

[159] Z. Chen, K. Gryllias, and W. Li, "Intelligent fault diagnosis for rotary machinery using transferable convolutional neural network," *IEEE Trans. Ind. Informat.*, vol. 16, no. 1, pp. 339–349, Jan. 2020.

[160] H. Ren et al., "Time-series anomaly detection service at microsoft," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2019, pp. 3009–3017.

[161] Q. Guo, Y. Li, Y. Song, D. Wang, and W. Chen, "Intelligent fault diagnosis method based on full 1-D convolutional generative adversarial network," *IEEE Trans. Ind. Informat.*, vol. 16, no. 3, pp. 2044–2053, Mar. 2020.

[162] N. K. Thanigaivelan, E. Nigussie, R. K. Kanth, S. Virtanen, and J. Isoaho, "Distributed internal anomaly detection system for Internet-of-Things," in *Proc. 13th IEEE Annu. Consum. Commun. Netw. Conf. (CCNC)*, 2016, pp. 319–320.

[163] X. Li, W. Zhang, Q. Ding, and X. Li, "Diagnosing rotating machines with weakly supervised data using deep transfer learning," *IEEE Trans. Ind. Informat.*, vol. 16, no. 3, pp. 1688–1697, Mar. 2020.

[164] C. Chen, F. Shen, J. Xu, and R. Yan, "Probabilistic latent semantic analysis-based gear fault diagnosis under variable working conditions," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 6, pp. 2845–2857, Jun. 2020.

[165] T. Ince, S. Kiranyaz, L. Eren, M. Askar, and M. Gabbouj, "Real-time motor fault detection by 1-D convolutional neural networks," *IEEE Trans. Ind. Electron.*, vol. 63, no. 11, pp. 7067–7075, Nov. 2016.

[166] X. Li, W. Zhang, N.-X. Xu, and Q. Ding, "Deep learning-based machinery fault diagnostics with domain adaptation across sensors at different places," *IEEE Trans. Ind. Electron.*, vol. 67, no. 8, pp. 6785–6794, Aug. 2020.

[167] Z. Ren, W. Zhang, and Z. Zhang, "A deep nonnegative matrix factorization approach via autoencoder for nonlinear fault detection," *IEEE Trans. Ind. Informat.*, vol. 16, no. 8, pp. 5042–5052, Aug. 2020.

[168] D. Adhikari et al., "A comprehensive survey on imputation of missing data in Internet of Things," *ACM Comput. Surveys*, vol. 55, no. 7, p. 133, 2022.

[169] Y. Li and L. E. Parker, "A spatial-temporal imputation technique for classification with missing data in a wireless sensor network," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2008, pp. 3272–3279.

[170] M. Halatchev and L. Gruenwald, "Estimating missing values in related sensor data streams," in *Proc. COMAD*, 2005, pp. 83–94.

[171] M. Dzaferagic, N. Marchetti, and I. Macaluso, "Fault detection and classification in industrial IoT in case of missing sensor data," *IEEE Internet Things J.*, vol. 9, no. 11, pp. 8892–8900, Jun. 2022.

[172] M. Anbarasan et al., "Detection of flood disaster system based on IoT, big data and convolutional deep neural network," *Comput. Commun.*, vol. 150, pp. 150–157, Jan. 2020.

[173] A. C. Harvey and R. G. Pierse, "Estimating missing observations in economic time series," *J. Amer. Statist. Assoc.*, vol. 79, no. 385, pp. 125–131, 1984.

[174] N. Alavi, J. S. Warland, and A. A. Berg, "Filling gaps in evapotranspiration measurements for water budget studies: Evaluation of a kalman filtering approach," *Agricul. Forest Meteorol.*, vol. 141, no. 1, pp. 57–66, 2006.

[175] S. Moritz and T. Bartz-Beielstein, "ImputeTS: Time series missing value imputation in R," *R J.*, vol. 9, no. 1, pp. 207–218, 2017.

[176] A. M. Moffat et al., "Comprehensive comparison of gap-filling techniques for eddy covariance net carbon fluxes," *Agricul. Forest Meteorol.*, vol. 147, nos. 3–4, pp. 209–232, 2007.

[177] G. H. Lee, J. Han, and J. K. Choi, "MPdist-based missing data imputation for supporting big data analyses in IoT-based applications," *Future Gener. Comput. Syst.*, vol. 125, pp. 421–432, Dec. 2021.

[178] M. Lee, J. An, and Y. Lee, "Missing-value imputation of continuous missing based on deep imputation network using correlations among multiple IoT data streams in a smart space," *IEICE Trans. Inf. Syst.*, vol. 102, no. 2, pp. 289–298, 2019.

[179] N. U. Okafor and D. T. Delaney, "Missing data imputation on IoT sensor networks: Implications for on-site sensor calibration," *IEEE Sensors J.*, vol. 21, no. 20, pp. 22833–22845, Oct. 2021.

[180] R. Vangipuram, R. K. Gunupudi, V. K. Puligadda, and J. Vinjamuri, "A machine learning approach for imputation and anomaly detection in IoT environment," *Expert Syst.*, vol. 37, no. 5, p. e12556, 2020.

[181] A. González-Vidal, P. Rathore, A. S. Rao, J. Mendoza-Bernal, M. Palaniswami, and A. F. Skarmeta-Gómez, "Missing data imputation with bayesian maximum entropy for Internet of Things applications," *IEEE Internet Things J.*, vol. 8, no. 21, pp. 16108–16120, Nov. 2021.

[182] X. Yan, W. Xiong, L. Hu, F. Wang, and K. Zhao, "Missing value imputation based on Gaussian mixture model for the Internet of Things," *Math. Probl. Eng.*, vol. 2015, Mar. 2015, Art. no. 548605.

[183] Z. Ding, G. Mei, S. Cuomo, Y. Li, and N. Xu, "Comparison of estimating missing values in IoT time series data using different interpolation algorithms," *Int. J. Parallel Program.*, vol. 48, no. 3, pp. 534–548, 2020.

[184] R. Karkare, R. Paffenroth, and D. Apelian, "Self-supervised deep hadamard autoencoders for treating missing data: A case study in manufacturing," *Integr. Mater. Manuf. Innovat.*, vol. 11, pp. 187–197, Mar. 2022.

[185] İ. Kök and S. Özdemir, "DeepMDP: A novel deep-learning-based missing data prediction protocol for IoT," *IEEE Internet Things J.*, vol. 8, no. 1, pp. 232–243, Jan. 2021.

[186] C. Velasco-Gallego and I. Lazakis, "A novel framework for imputing large gaps of missing values from time series sensor data of marine machinery systems," *Ships Offshore Structures*, vol. 17, no. 8, pp. 1802–1811, 2022.

[187] B. Venkatesh and J. Anuradha, "A review of feature selection and its methods," *Cybern. Inf. Technol.*, vol. 19, no. 1, pp. 3–26, 2019.

[188] H. Peng, F. Long, and C. Ding, "Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 8, pp. 1226–1238, Aug. 2005.

[189] P. A. Estévez, M. Tesmer, C. A. Perez, and J. M. Zurada, "Normalized mutual information feature selection," *IEEE Trans. Neural Netw.*, vol. 20, no. 2, pp. 189–201, Feb. 2009.

[190] Q. Jiang and X. Yan, "Monitoring multi-mode plant-wide processes by using mutual information-based multi-block pca, joint probability, and bayesian inference," *Chemometr. Intell. Lab. Syst.*, vol. 136, pp. 121–137, Aug. 2014.

[191] Q. Jiang and X. Yan, "Plant-wide process monitoring based on mutual information–multiblock principal component analysis," *ISA Trans.*, vol. 53, no. 5, pp. 1516–1527, 2014.

[192] J. Huang and X. Yan, "Quality relevant and independent two block monitoring based on mutual information and KPCA," *IEEE Trans. Ind. Electron.*, vol. 64, no. 8, pp. 6518–6527, Aug. 2017.

[193] K. Ghosh, M. Ramteke, and R. Srinivasan, "Optimal variable selection for effective statistical process monitoring," *Comput. Chem. Eng.*, vol. 60, pp. 260–276, Jan. 2014.

[194] S. M. Namburu, M. S. Azam, J. Luo, K. Choi, and K. R. Pattipati, "Data-driven modeling, fault diagnosis and optimal sensor selection for HVAC chillers," *IEEE Trans. Autom. Sci. Eng.*, vol. 4, no. 3, pp. 469–473, Jul. 2007.

[195] Q. Jiang, X. Yan, and B. Huang, "Performance-driven distributed PCA process monitoring based on fault-relevant variable selection and Bayesian inference," *IEEE Trans. Ind. Electron.*, vol. 63, no. 1, pp. 377–386, Jan. 2016.

[196] Z. Ge and J. Chen, "Plant-wide industrial process monitoring: A distributed modeling framework," *IEEE Trans. Ind. Informat.*, vol. 12, no. 1, pp. 310–321, Feb. 2016.

[197] Z. Ge and Z. Song, "Distributed PCA model for plant-wide process monitoring," *Ind. Eng. Chem. Res.*, vol. 52, no. 5, pp. 1947–1957, 2013.

[198] Y. Zhang and C. Ma, "Decentralized fault diagnosis using multiblock kernel independent component analysis," *Chem. Eng. Res. Design*, vol. 90, no. 5, pp. 667–676, 2012.

[199] C. Tong and X. Yan, "A novel decentralized process monitoring scheme using a modified multiblock PCA algorithm," *IEEE Trans. Autom. Sci. Eng.*, vol. 14, no. 2, pp. 1129–1138, Apr. 2017.

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[200] J. Zhu, Z. Ge, and Z. Song, "Distributed parallel PCA for modeling and monitoring of large-scale plant-wide processes with big data," *IEEE Trans. Ind. Informat.*, vol. 13, no. 4, pp. 1877–1885, Aug. 2017.
[201] Z. Chen et al., "A distributed canonical correlation analysis-based fault detection method for plant-wide process monitoring," *IEEE Trans. Ind. Informat.*, vol. 15, no. 5, pp. 2710–2720, May 2019.
[202] C. Tong, Y. Song, and X. Yan, "Distributed statistical process monitoring based on four-subspace construction and bayesian inference," *Ind. Eng. Chem. Res.*, vol. 52, no. 29, pp. 9897–9907, 2013.
[203] D. Li, Y. Zhou, G. Hu, and C. J. Spanos, "Optimal sensor configuration and feature selection for AHU fault detection and diagnosis," *IEEE Trans. Ind. Informat.*, vol. 13, no. 3, pp. 1369–1380, Jun. 2017.
[204] P. Strauß, M. Schmitz, R. Wöstmann, and J. Deuse, "Enabling of predictive maintenance in the brownfield through low-cost sensors, an IIoT-architecture and machine learning," in *Proc. IEEE Int. Conf. Big Data*, 2018, pp. 1474–1483.
[205] O. Gungor, T. S. Rosing, and B. Aksanli, "DOWELL: Diversity-induced optimally weighted ensemble learner for predictive maintenance of Industrial Internet of Things devices," *IEEE Internet Things J.*, vol. 9, no. 4, pp. 3125–3134, Feb. 2022.
[206] M. Fernandes, A. Canito, V. Bolón-Canedo, L. Conceição, I. Praça, and G. Marreiros, "Data analysis and feature selection for predictive maintenance: A case-study in the metallurgic industry," *Int. J. Inf. Manag.*, vol. 46, pp. 252–262, Jun. 2019.
[207] O. O. Aremu, R. A. Cody, D. Hyland-Wood, and P. R. McAree, "A relative entropy based feature selection framework for asset data in predictive maintenance," *Comput. Ind. Eng.*, vol. 145, Jul. 2020, Art. no. 106536.
[208] R. D. Pandey and I. Snigdh, "Validity as a measure of data quality in Internet of Things systems," *Wireless Pers. Commun.*, vol. 126, pp. 933–948, Jun. 2022.

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